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CENG 408 – FINAL REPORT

Project Title:

**AI-Supported Lung Nodule Detection and
Classification Using Medical Imaging**

Project Name:

LungXai

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Abstract

Lung cancer is one of the leading causes of cancer-related mortality worldwide, and early detection of malignant pulmonary nodules plays a critical role in improving patient outcomes. Although low-dose computed tomography (LDCT) enables early identification of lung nodules, manual assessment remains challenging due to large scan volumes, subtle visual differences between benign and malignant lesions, and inter-observer variability among radiologists. These challenges motivate the development of reliable and interpretable artificial intelligence–based decision support systems.

In this study, we present LungXai, a modular deep learning framework for lung nodule malignancy classification using chest CT images. The system adopts a 2.5D input representation, in which a small stack of adjacent axial slices is treated as a multi-channel input to a two-dimensional convolutional neural network. This approach captures limited volumetric context while maintaining computational efficiency. A ResNet-18 architecture is employed as a baseline classifier within the MONAI medical imaging ecosystem, ensuring standardized preprocessing, reproducibility, and extensibility.

The proposed pipeline processes raw DICOM images and XML-based radiologist annotations from the LIDC-IDRI dataset to construct annotation-guided 2.5D samples. Experimental evaluation demonstrates that meaningful discrimination between benign and malignant nodules can be achieved even with a lightweight baseline model. The analysis further highlights that ROC-AUC is a more reliable evaluation metric than accuracy in this imbalanced medical classification setting. In addition, a prototype user interface is developed to visualize model outputs and demonstrate practical system integration.

Rather than presenting a fully coupled end-to-end clinical solution, LungXai follows a baseline-first and progressive development strategy. More advanced extensions are intentionally deferred to later stages, allowing the current work to establish a reproducible and extensible foundation for anatomically grounded and interpretable lung nodule analysis.

1. Introduction

Lung cancer remains one of the leading causes of cancer-related mortality worldwide [1]. Although low-dose computed tomography (LDCT) enables the early detection of pulmonary nodules, clinical evaluation remains challenging due to the large number of slices per scan, subtle visual differences between benign and malignant nodules, and inter-observer variability in malignancy assessment. These challenges motivate the development of computer-aided decision support systems that can assist radiologists by providing consistent and reproducible predictions.

In recent years, deep learning-based approaches have become widely adopted for pulmonary nodule analysis. However, medical imaging introduces domain-specific constraints that are not typically encountered in natural-image applications, including heterogeneous voxel spacing, intensity calibration in Hounsfield Units (HU), and the use of specialized data formats such as DICOM and XML-based annotations. Addressing these constraints requires a controlled and standardized experimental pipeline.

In this study, we present **LungXai**, a modular research-oriented framework for lung nodule malignancy classification using chest CT images. The current phase of the project focuses on establishing a reliable baseline classification pipeline based on (i) a 2.5D input representation constructed from adjacent axial slices, (ii) a ResNet-18 classification model, and (iii) a prototype user interface (UI) for visualizing model outputs. The system is implemented using the MONAI framework to support standardized preprocessing and reproducible experimentation.

Rather than introducing a fully integrated end-to-end clinical system, this work follows a baseline-first and progressive development strategy. More advanced components - such as segmentation-assisted region-of-interest refinement, stronger classification backbones, and explainable artificial intelligence (XAI) techniques - are outside the scope of the current implementation and are not included in this phase.

2. Previous Studies

2.1 Previous Studies on Deep Learning–Based Lung Nodule Analysis

This section reviews prior studies related to deep learning–based lung nodule classification, focusing on the evolution of model architectures, comparative performance against clinical standards, the role of contextual information and explainability, and the overall progression of research in this domain.

2.1.1 Evolution of Deep Learning Architectures

Deep learning (DL) has fundamentally transformed medical image interpretation, replacing traditional radiomics pipelines that relied on handcrafted features with data driven Convolutional Neural Networks (CNNs). Early DL approaches utilized 2D CNNs, but these lacked volumetric awareness across contiguous CT slices [3]. This limitation led to the adoption of 3D CNNs as the standard for volumetric tasks like lung nodule detection. However, 3D convolution presents significant practical barriers, including cubic computational cost and the requirement for extensive annotated datasets, scaling poorly for large CT volumes [2].

To overcome the tradeoff between volumetric context and computational efficiency, researchers introduced 2.5D architectures. These models utilize a small stack of consecutive slices (multi-slice context) as input to a 2D network, effectively capturing local 3D context with reduced complexity [2]. Comparative analyses confirm the efficacy of this approach, showing that 2.5D DenseNet models can achieve a diagnostic performance comparable to that of thoracic radiologists [2]. Furthermore, a 2.5D DenseNet model demonstrated superior overall performance and discrimination, achieving a

significantly higher Area Under the Curve (AUC) of 0.921 compared to a 3D DenseNet (AUC 0.835) and a size based logistic model (AUC 0.836) in differentiating invasive pulmonary adenocarcinomas [2]. The 2.5D approach also showed a high specificity of 88.2% at 90% sensitivity [2].

2.1.2 DL Performance Compared to Clinical Standards

Recent systematic reviews and meta analyses validate the superior performance of DL-based models in predicting the risk of malignancy in CT-detected pulmonary nodules compared to conventional clinical tools [6]. A comprehensive meta analysis across seventeen externally validated studies (comprising 9,884 nodules) yielded the following key findings [6]:

- **DL vs. Physician Judgement:** DL-based models demonstrated a 1.6% increase in sensitivity compared to physician judgement alone [6]. While specificity was similar (pooled DL specificity 0.77 vs. physician 0.81), DL models showed superior pooled AUC of 0.86 compared to physician readers alone (AUC 0.83) [6].
- **DL vs. Clinical Risk Models:** DL models were 14.5% more sensitive and 7.4% more specific than clinical risk models (such as the Brock and Mayo models) alone [6]. The pooled AUC for DL models (0.86) was significantly superior to clinical risk models (AUC 0.79) [6].

This evidence suggests that DL-based models, such as DenseNet-121, which achieved an accuracy of 90.39% and specificity of 93.65% in comparative studies, are already justified for routine deployment alongside experienced physicians [4, 6].

2.1.3 The Role of Nodule Context and Explainability

While CNNs are effective in feature extraction, studies indicate that high performance often relies on hybrid architectures and considering the nodule's surrounding environment [3, 5].

- **Impact of Fibrotic Microenvironment:** Novel research has demonstrated that incorporating the surrounding pulmonary context, particularly pulmonary fibrosis (fibrotic microenvironment), significantly improves nodule malignancy classification [5]. A 3D Attention Gated Network (3D AG-Net) trained with semantic fibrosis metadata achieved the best results with 80.84% accuracy and 0.89 AUC [5]. This study confirmed that models trained with both the nodule and its microenvironment outperformed models trained on the nodule alone, with the ability to detect malignant nodules increasing by 9.21% when fibrosis data was presented [5].
- **Need for Hybrid Models and XAI:** The conventional convolution process may be

ineffective for feature extraction alone, leading to the development of hybrid models utilizing attention blocks to overcome these limitations [3]. Furthermore, studies focusing on Explainable AI (XAI) emphasize that predictive estimates are sometimes not understood or interpreted by clinicians [3]. The development of visualizable models, such as the 3D AG-Net, which uses attention gates to filter and visualize features at different network depths, provides crucial interpretability that is highly valued in clinical settings [5].

In conclusion, the literature points toward the efficacy of 2.5D and attention based architectures for achieving high performance with computational efficiency [2], and highlights the necessity of integrating these systems into a standardized, clinically translatable workflow (e.g., using frameworks like MONAI) that can handle complex data constraints and provide the required diagnostic accuracy and explainability [2, 6, 5].

2.1.4 Evolution of Research from Past to Present

Research on AI-based lung nodule analysis has evolved from early 2D CNN approaches toward methods that better capture volumetric context and clinical usability. While 2D models are computationally efficient, their inability to represent inter-slice continuity motivated 3D CNNs, which improved spatial awareness but introduced substantial computational cost. To balance these trade-offs, 2.5D architectures emerged as a practical compromise by stacking adjacent slices as multi-channel input, capturing local 3D context with 2D-level efficiency. More recently, attention mechanisms and explainability techniques (e.g., Grad-CAM) have gained importance to address the clinical “black box” concern, while unified ecosystems such as MONAI have become central by standardizing preprocessing, training, evaluation, and deployment pathways for real-world medical imaging workflows.

2.2 Overview of Existing Frameworks

The nnU-Net framework introduced automated configuration for segmentation pipelines, standardizing preprocessing, architecture selection, and training parameters. It has achieved state of the art results in organ segmentation challenges but is task specific and lacks built in modules for classification, explainability, or deployment. Integrating nnU-Net into classification workflows typically requires additional frameworks.

TorchIO focuses on preprocessing and augmentation, enabling advanced data transformations

such as elastic deformations, bias field simulation, and random noise. Although highly useful for dataset preparation, TorchIO does not provide model architectures, evaluation metrics, or deployment infrastructure.

Both nnU-Net and TorchIO contribute valuable utilities; however, they function as partial solutions. In contrast, MONAI integrates multiple components commonly required in medical AI research within a single coherent ecosystem. For a project that aims to progress from research to clinically usable outcomes, such integration ensures maintainability, reproducibility, and compliance with hospital IT standards.

2.3 MONAI: The Unified Medical AI Ecosystem

The Medical Open Network for AI (MONAI), founded jointly by NVIDIA and King's College London, represents a purpose built ecosystem for medical imaging research and deployment. It is fully open source (Apache 2.0 license) and implemented in Python using the PyTorch backend. MONAI's modular architecture consists of several tightly integrated subsystems.

The `monai.transforms` module provides domain specific image transformations, including intensity normalization, resampling, cropping, random affine transformation, and Gaussian noise addition. The `monai.networks` library contains deep learning architectures optimized for medical imaging, such as DenseNet, UNet, AttentionUNet, and SwinUNETR, supporting 2D, 2.5D, and 3D inputs. The `monai.metrics` module implements standardized evaluation metrics including Dice coefficient, Hausdorff distance, AUC, and FROC. The `monai.inferers` package enables efficient sliding window and patchbased inference for large 3D volumes, while `monai.visualize` provides utilities for Grad-CAM, feature maps, and training visualizations. Finally, `monai.apps` includes pre trained models and ready to use pipelines for segmentation, classification, and detection tasks.

MONAI's data pipeline supports flexible input dimensionality. Through its composable transforms, multiple adjacent CT slices can be concatenated along the channel dimension to simulate a 2.5D stack. The framework's `CacheDataset` and `SmartCacheDataset` classes allow efficient on thefly augmentation of slice groups, while `sliding_window_inference` enables inference on large 3D scans by processing 2.5D patches sequentially—an efficient hybrid strategy between 2D and 3D.

Furthermore, MONAI provides native support for explainable AI via Grad-CAM and Integrated Gradients, allowing visualization of malignancy driving regions directly on CT slices. Deployment-oriented components such as MONAI Deploy are considered future work and are not part of the current implementation.

Unlike modular utilities such as TorchIO or nnU-Net MONAI supports a unified workflow from preprocessing to inference within a single standardized environment.” This eliminates compatibility issues between libraries, reduces code redundancy, and improves experiment traceability for research purposes. Therefore, the current study exclusively adopts MONAI for all stages of development[7]. Table 2.3.1 summarizes the key differences between MONAI, nnU-Net, and TorchIO, highlighting MONAI’s advantage as a unified end-to-end medical AI framework.

Framework Comparison

Table 2.3.1: Comparison of MONAI, nnU-Net, and TorchIO frameworks in terms of functionality, flexibility, and deployment support.

Criterion	MONAI	nnU-Net	TorchIO
Goal	An end-to-end framework for medical imaging and deep learning.	Self-adapting automatic segmentation system.	A library for medical image preparation and augmentation.
Input Formats	DICOM, NIFTI, PNG,JPG,3D volumetric data.	NIFTI heavy.	NIFTI, DICOM, etc.; focus: preprocessing.
Preprocessing Features	Medical image normalization, spacing adjustment, intensity transforms, lung windowing, DenseNet support.	Automatic voxel spacing adjustment, normalization, resampling.	Built in loading, non rigid transforms, random intensity, bias field noise, elastic deformation.

Model Architecture	Systems supporting 2D/ 3D U-Net, SegResNet, DenseNet, Swin-UNETR, Vision Transformer based systems.	2D/3D UNet/3D cascade UNet automatic configuration.	Model is robust, users add their own models.
Core Feature	Education -	Automatic	Modern augmentation and
	Assessment - XAI - Inference - Clinical deploy pipeline.	hyperparameter selection and preprocessing.	data pipeline control.
2D-3D Support	Native 2D and 3D pipeline.	2D, 3D, 3D cascade.	2D & 3D
Explainability (XAI)	GRAD-CAM, IG, SHAP integration.	Not natively present; external.	No
Annotation / GUI	MONAI Label (3D Slicer / OHIF support).	No annotation support.	Annotation support.
Hospital / PACS Integration	Yes (MONAI Deploy DICOM, HL7, FHIR).	No	No
Flexibility	Segmentation + classification + detection + registration.	Only segmentation.	Preprocessing library for every model type.
Optimization Strategy	Customizable loss functions, mixed precision, distributed training.	Fully automatic model selection and hyperparameter optimization.	Data pipeline acceleration (patch based training support).
Advantage	Complete ecosystem supporting research to clinical deployment.	SOTA benchmark in segmentation.	Ready made modern medical augmentation set.
Limitation	High learning curve.	Preprocessing and segmentation only.	Does not include model training.

Commercial Clinical AI Systems Clinical deployment has been exemplified by Qure.ai and Optellum. Qure.ai's qCT-LN Quant automates volumetric nodule quantification and serial growth tracking, integrating into PACS and providing structured reports. Optellum's Virtual Nodule Clinic specializes in malignancy risk stratification, offering dashboards with color coded risk scores and follow up management tools[8][9].

Both systems are FDA-cleared, but their proprietary architectures prevent academic reproducibility. These products illustrate the importance of combining high algorithmic performance with interpretability and workflow usability core design principles also emphasized in our system. Table 2.1.1 provides a comparative overview of Qure.ai and Optellum Virtual Nodule Clinic, highlighting differences in clinical focus, output structure, and workflow integration.

Commercial System Comparison

Table 2.4.1: Comparison of Qure.ai and Optellum Virtual Nodule Clinic in terms of clinical functionality and workflow integration.

Criterion	Qure.ai	Optellum Virtual Nodule Clinic
Medical Use Area	Automatic findings and reports for Chest X-ray and brain CT.	Malignancy risk assessment of lung nodules.
Data Input	X-ray, CT.	CT (thorax).
Output Structure	Automatic report draft, finding annotation, heatmap.	Nodule based risk score, case management panel.
Clinical Workflow Model	Triage + automatic reporting.	Risk stratification + clinical decision support.
Interface	Report editor + visual finding annotation.	Nodule list + malignancy risk matrix + follow up plan.
Performed Task	Automatic writing of findings, decision acceleration.	Clinical probability support for nodule malignancy.

Decision Authority	Physician approved AI recommendation.	Multidisciplinary team assessment.
Most Prominent Feature	Automatic radiology report	Nodule malignancy scoring and patient specific risk planning.
Strength	Operational speed and report standardization.	Risk modeling close to clinical decision.
Limitation	Not focused on nodule based malignancy.	Area of operation is lung nodules only.

3. Research Questions and Approach

Q1. Why is a 2.5D hybrid MONAI approach preferred over purely 2D or 3D CNN architectures?

2D models require significantly less GPU memory and offer faster training times, making them ideal for limited computational environments. However, they lack volumetric contextual awareness. 3D models process entire volumes, improving spatial understanding but demanding large VRAM and prolonged training.

A 2.5D hybrid approach implemented in MONAI provides the optimal balance stacking multiple consecutive CT slices as input channels to a 2D CNN. This configuration captures local 3D context while maintaining training efficiency.

Q2. Which deep learning architectures are most suitable for segmentation and classification tasks in lung nodule malignancy assessment?

In the current phase, a lightweight ResNet-18 model is adopted to establish a stable and reproducible 2.1 D baseline under limited hardware resources. In subsequent phases, stronger feature extractors will be explored for comparative evaluation.

- **Segmentation (planned):** architectures such as SegResNet and U-Net will be evaluated for robust mask generation and ROI refinement.
- **Classification (planned extensions):** architectures such as DenseNet121, ResNet50,

and EfficientNet variants will be investigated to improve discrimination performance and enable richer representational capacity.

- **Ensemble (optional future work):** combinations of complementary backbones may be explored to balance performance, robustness, and interpretability.

Q3. How will Explainable AI (XAI) techniques be implemented within the system?

Explainability is considered an important extension for clinical interpretability; however, XAI methods are not integrated in the current implementation. The planned integration strategy and candidate techniques are presented in Section 4.2, while an exploratory reliability analysis is reported in Appendix A. surrounding structures. At the current stage, these techniques are not implemented and are considered part of planned future work, following the integration of segmentation and ROI-guided learning.

Q4. How do hardware limitations affect model selection, training, and evaluation strategies?

Hardware constraints directly influenced model selection and experimental design. The current system prioritizes a computationally efficient 2.5D ResNet-18 baseline to validate the data pipeline and training workflow under limited GPU resources. Accordingly, training and evaluation settings were chosen to preserve stability and reproducibility while enabling iterative experimentation.

4. Proposed Contributions

This section presents the studies carried out within the scope of the LungXai project and the outputs obtained. The contributions are organized based on the current stage of development of the project, with completed implementations and planned future studies addressed under separate headings. This structure aims to systematically present the scope of the work and its development process.

4.1 Implemented Contributions (Current Phase)

4.1.1 ROI-Based End-to-End 2.5D Preprocessing and Data Preparation Pipeline

In the current phase of this study, an end-to-end preprocessing pipeline has been developed to transform raw CT and SEG data from the LIDC-IDRI dataset into ROI-based 2.5D samples suitable for training. The pipeline includes CT–SEG matching, full-volume mask reconstruction, ROI extraction, 2.5D sample generation, and preprocessing cache creation.

Within the system, CT and SEG matching is performed using UID-based correspondence, and SEG frames are mapped into CT slice space to generate anatomically aligned full-volume masks. Afterward, bounding-box-based ROI extraction is applied around the nodule region, and the extracted ROIs are resized to a standardized input resolution.

The pipeline also includes:

- positive-center-only sampling,
- patient-level split,
- and .npz-based preprocessing cache infrastructure.

This preprocessing framework provides a shared data foundation not only for segmentation, but also for classification, Grad-CAM, and reporting modules.

4.1.2 Implementation of the SegResNet-Based Segmentation Pipeline

A SegResNet-based segmentation pipeline has been implemented to achieve anatomically meaningful nodule localization. This structure was trained on ROI-based 2.5D input samples to learn nodule masks.

The segmentation pipeline includes:

- training and validation Dice analysis,
- probability map generation,
- checkpoint-based model saving,
- and qualitative validation outputs.

The experimental results demonstrated that the model was capable of learning meaningful spatial representations. Validation examples showed that the model successfully localized the main nodule region in most cases. Probability map analysis further indicated that high-activation regions were generally concentrated around the lesion area.

This structure provides a foundation for future segmentation-guided classification and anatomical validation studies.

4.1.3 Knowledge Distillation-Based Classification Structure

A teacher–student knowledge distillation framework was developed for malignancy classification. In this structure, a high-capacity 3D teacher model learns volumetric nodule-related information, while a lighter 2.5D student model learns from tri-planar axial, coronal, and sagittal representations.

The classification pipeline includes:

- preprocessing,

- patch extraction,
- candidate-centered sample generation,
- distillation loss computation,
- and ROC-AUC-based evaluation.

Experimental findings showed that the student model improved under teacher supervision. Positive trends were observed in both accuracy and distillation loss throughout training.

In addition, ROC-AUC was found to be more informative than accuracy for this medical classification problem and was therefore adopted as the primary evaluation metric for model selection.

4.1.4 Grad-CAM-Based Interpretability (XAI) Module

A Grad-CAM-based interpretability module was integrated into the system to visually inspect model decisions. This module generates heatmaps highlighting the spatial regions influencing model predictions.

The Grad-CAM pipeline includes:

- inference-ready preprocessing,
- activation map generation,
- heatmap overlay operations,
- and confidence analysis.

The generated heatmaps showed that model attention was generally concentrated around the nodule region, improving the inspectability of the system's decision-making process.

However, some activation overflow behavior was observed in small nodules, indicating that future quantitative Grad-CAM validation studies are necessary.

4.1.5 Anatomical Nodule Localization Module

An anatomical localization algorithm based on the `estimate_candidate_location` function was implemented to estimate the anatomical positions of nodules.

This structure utilizes:

- nodule center coordinates,
- DICOM metadata,
- `ImagePositionPatient`,
- `ImageOrientationPatient`,
- and slice-based lung position analysis

to estimate the anatomical region of each candidate nodule.

The system performs:

- right/left lung estimation,
- upper/middle/lower lobe estimation,
- slice fraction computation,
- and localization confidence generation.

As a result, the system can produce anatomical labels such as:

- RUL,
- RML,
- RLL,
- LUL,
- and LLL.

This localization module provides a foundation for automated report generation and UI-based visual marking systems.

4.1.6 Turkish NLP and Rule-Based Clinical Reporting System

A Turkish automatic report generation system based on the Fleischner 2017 guidelines was developed within the project. This structure generates clinical follow-up recommendations using nodule size, risk factors, and anatomical localization information together with predefined rule-based clinical pathways.

The system includes:

- risk-based follow-up recommendations,
- multiple nodule management support,
- and fallback-based safe reporting mechanisms.

In addition, through BERTurk-based NLP integration, supportive information such as:

- smoking history,
- malignancy-related risk signals,
- and morphological expressions

can be extracted from Turkish clinical notes.

The NLP outputs are stored as supportive auxiliary fields rather than directly controlling the final clinical recommendation. The primary recommendation mechanism remains fully deterministic and rule-based.

4.1.7 Computational Optimization and Cache Infrastructure

An .npz-based cache infrastructure was developed to reduce preprocessing overhead within the pipeline. This structure enables preprocessing outputs to be reused and prevents repeated CT-SEG reconstruction during every training epoch.

This optimization:

- reduced preprocessing time,
- minimized GPU idle time,
- and improved training stability.

In addition, performance-related parameters such as:

- batch size,
- preprocessing throughput,
- memory usage,
- and validation frequency

were monitored to support future scalability toward larger segmentation and multitask learning frameworks.

4.1.8 Partial Implementation of the User Interface (UI) Layer

A user interface (UI) layer was developed to support visualization of system outputs and improve user interaction.

The current UI supports:

- input selection,
- model inference execution,
- classification result visualization,
- Grad-CAM heatmap visualization,
- and report output presentation.

Although the current structure remains at the prototype level, it establishes a foundational clinical interaction layer that can later be extended with:

- segmentation outputs,
- anatomical localization results,
- and additional explainability visualizations.

5. Methodology and System Approach

5.1 Distinctive Aspects of the Study

This study adopts a multi-stage AI-assisted lung nodule analysis framework that combines preprocessing, segmentation, classification, explainability, anatomical localization, Turkish NLP support, and user interface integration within a single workflow. Unlike many studies that focus only on image classification, the LungXai framework aims to establish an anatomically aware and interpretable pipeline extending from raw DICOM processing to clinical report generation.

The current system includes the following components:

- ROI-based 2.5D preprocessing pipeline,
- UID-based CT–SEG matching and full-volume mask reconstruction,
- SegResNet-based segmentation,
- knowledge distillation-based classification,
- Grad-CAM-based interpretability,
- anatomical nodule localization,
- Turkish NLP-assisted clinical reporting,
- preprocessing cache infrastructure,
- and UI-assisted visualization.

The main objectives of this system design are:

- to establish anatomically consistent preprocessing and learning stages,
- to support interpretable AI-assisted decision-making,
- to evaluate segmentation and classification behavior together,
- to reduce background influence through ROI-focused processing,
- and to create a modular framework suitable for future clinical deployment and system expansion.

Unlike single-stage classification pipelines, the proposed framework integrates multiple complementary modules while preserving experimental traceability and reproducibility.

5.2 Dataset Description: LIDC-IDRI

The Lung Image Database Consortium and Image Database Resource Initiative (LIDC-IDRI) dataset is one of the most comprehensive publicly available thoracic CT datasets used for pulmonary nodule analysis, segmentation, malignancy classification, and explainable medical AI research.

The dataset consists of 1,018 patient studies containing:

- full chest CT scans stored in DICOM format,
- XML-based radiologist annotations,
- and segmentation-related metadata.

Each study includes:

- axial CT volumes,
- nodule contour annotations,
- malignancy ratings,
- and additional semantic radiological characteristics.

The variability in acquisition parameters, slice thickness, and spacing reflects real-world clinical heterogeneity and improves the dataset’s suitability for robust AI experimentation.

5.2.1 Nodule Annotations and SEG Data

Radiologists independently annotated each CT volume following a multi-reader protocol. For each annotated nodule, the dataset provides:

- slice-wise contours,
- three-dimensional centroid coordinates,
- malignancy scores,
- and semantic radiological attributes.

The project utilizes both XML-based annotations and DICOM SEG-related structures during preprocessing and validation.

Annotated nodules are categorized into:

- nodules ≥ 3 mm,
- and nodules < 3 mm.

Only nodules ≥ 3 mm are included in this study to ensure sufficient visual and structural information for segmentation and classification learning.

5.2.2 Malignancy Label Definition

The malignancy score in the LIDC-IDRI dataset ranges from 1 to 5.

The labels are interpreted as:

- Scores 1–2 $\rightarrow$ benign,
- Score 3 $\rightarrow$ indeterminate,
- Scores 4–5 $\rightarrow$ malignant.

Since score 3 represents diagnostic uncertainty, these samples are excluded to reduce label noise and improve binary classification reliability.

To reduce inter-reader variability, malignancy scores assigned by multiple radiologists are aggregated before final label assignment.

5.2.3 ROI-Based Sample Definition

Instead of treating individual slices independently, the proposed framework constructs ROI-based 2.5D samples centered around the annotated nodule region.

The preprocessing pipeline includes:

- CT–SEG matching,
- full-volume mask reconstruction,
- ROI extraction,
- resizing,
- and five-slice 2.5D generation.

Each generated sample contains:

- neighboring contextual slices,
- the center slice,
- and anatomically aligned mask information.

This representation allows the model to preserve local volumetric context while maintaining lower computational cost compared to fully 3D processing.

5.2.4 Segmentation Preprocessing and ROI Generation

To prepare segmentation-ready inputs, SEG frames are reconstructed into CT slice space after UID-based CT–SEG matching. Following mask reconstruction, a bounding-box-based ROI extraction strategy is applied around the lesion region with a fixed margin.

The extracted ROIs are resized into a standardized resolution and converted into five-channel 2.5D representations using:

- $[k-2, k-1, k, k+1, k+2]$

neighboring slice structures.

A positive-center-only sampling policy is used to ensure that the center slice contains the target lesion mask.

To improve preprocessing efficiency, generated image-mask pairs are stored as compact .npz cache files and reused during training.

5.2.5 Knowledge Distillation Sample Preparation

For the knowledge distillation-based classification structure, candidate nodule coordinates are converted from world coordinates into voxel coordinates.

The preprocessing stage includes:

- extraction of $32 \times 32 \times 32$ 3D patches,
- HU clipping,
- normalization,
- and tri-planar student input generation.

The teacher model processes full 3D patches, while the student model receives axial, coronal, and sagittal slices as a lightweight 2.5D representation.

This structure allows the student model to approximate volumetric information while maintaining lower computational requirements.

5.2.6 Anatomical Localization Strategy

The system includes a rule-based anatomical localization module implemented through the `estimate_candidate_location` function.

This algorithm:

- extracts candidate center coordinates,
- calculates patient-space coordinates using DICOM metadata,
- estimates right/left lung position,
- computes slice position inside the lung volume,
- and assigns anatomical region labels such as:
 - RUL,
 - RML,
 - RLL,
 - LUL,
 - and LLL.

The localization module additionally generates:

- slice fraction values,
- and localization confidence information.

These outputs are later used within the reporting and UI visualization pipeline.

5.3 Preprocessing Steps

Before training and inference, raw DICOM data undergo a standardized preprocessing workflow designed to ensure anatomical consistency and stable model training.

The preprocessing pipeline includes:

- DICOM loading,
- HU conversion,
- intensity normalization,
- CT–SEG matching,
- full-volume mask reconstruction,

- ROI extraction,
- 2.5D sample generation,
- normalization,
- and cache generation.

For classification preprocessing:

- candidate-centered patches are generated,
- tri-planar student inputs are created,
- and normalization is applied.

For Grad-CAM preprocessing:

- inference-ready normalized inputs are generated,
- and heatmap overlay compatibility is preserved.

These preprocessing stages establish a common anatomically consistent foundation for segmentation, classification, explainability, localization, and reporting modules.

5.4 Processing Pipeline

5.4.1 Current System Pipeline

The implemented LungXai processing pipeline is summarized as follows:

DICOM + XML + SEG Data

- Metadata inspection and CT–SEG matching
- Full-volume mask reconstruction
- ROI extraction and normalization
- 2.5D sample generation
- SegResNet segmentation
- Knowledge distillation-based classification
- Grad-CAM visualization
- Anatomical localization estimation
- Turkish NLP-assisted report generation
- UI-assisted visualization and output presentation

Unlike earlier baseline-only systems, the current framework integrates preprocessing, segmentation, classification, explainability, localization, and reporting components into a unified workflow.

5.5 Algorithms Used in the Current System

Table 5.5.1 summarizes the main algorithms and system components implemented in the current LungXai framework.

Table 5.5.1 Algorithms and system components implemented in the current LungXai framework

Component	Current Implementation
Input Representation	ROI-based 2.5D slice stacking
CT–SEG Matching	UID-based DICOM matching
Segmentation	SegResNet
Classification	Knowledge Distillation (3D Teacher → 2.5D Student)
Explainability	Grad-CAM
Anatomical Localization	estimate candidate location
Clinical Reporting	Fleischner-based rule engine
Turkish NLP	BERTurk-assisted auxiliary analysis
Optimization	.npz preprocessing cache
UI	Prototype visualization interface

5.6 Explainable Artificial Intelligence (XAI)

In medical image analysis, simply generating accurate predictions is not enough. Understanding which spatial regions influence model decisions is also crucial. Therefore, the LungXai framework integrates Grad-CAM-based explainability support by showing the section with the highest probability of nodule formation and the masked region, as determined from previous steps, using a heat map method.

The implemented Grad-CAM module generates class-relevant activation heatmaps by analyzing gradients flowing through the final convolutional layers of the model.

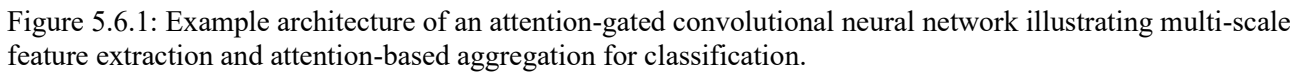
These heatmaps:

- highlight spatially relevant regions,
- improve interpretability,
- and support visual inspection of model behavior.

The generated activation maps are evaluated together with:

- raw CT slices,

- This structure helps determine whether the model focuses on anatomically meaningful nodule regions instead of unrelated background structures.



During training, multiple evaluation metrics were monitored across epochs.

- training Dice,
- validation Dice,
- training loss,

- and validation loss

were analyzed.

For classification:

- accuracy,
- ROC-AUC,
- and distillation loss

were monitored throughout training.

The experiments demonstrated that:

- segmentation Dice scores improved steadily,
- distillation loss decreased,
- and student classification accuracy improved under teacher supervision.

In addition, ROC-AUC was treated as the primary classification evaluation metric due to class imbalance considerations in medical datasets.

The generated validation outputs, probability maps, Grad-CAM visualizations, and localization results were further used for qualitative inspection of model behavior and anatomical consistency.

6 . System Architecture and Implementation

This section presents the end-to-end system architecture and implementation approach of the LungXai project. Rather than focusing on the theoretical aspects of the employed deep learning models, this section aims to describe how the system is structured, how its components interact, and how the overall workflow is managed from raw medical data to final outputs. The proposed architecture demonstrates that the project is not merely a model-centric experiment, but an engineering-oriented research framework designed with reproducibility and extensibility in mind.

6.1 Overall System Approach

LungXai is designed as a multi-stage decision support system for lung nodule analysis using chest computed tomography (CT) images. Instead of directly feeding raw data into the learning model, the system follows a clearly defined and controlled processing pipeline. This design choice improves experimental consistency and reduces uncertainties inherent in

clinical imaging data.

Conceptually, the system is composed of logically separated layers responsible for data acquisition, preprocessing, representation learning, modeling, evaluation, and output generation. Each layer is designed with a specific responsibility and interacts with other layers in a loosely coupled manner, enabling modular development and future scalability.

6.2 Input Data and Preprocessing Layer

In the first stage of the system, thoracic CT images are ingested in DICOM format. Since raw pixel intensities vary across scanners and acquisition protocols, these values are not directly suitable for learning-based analysis. Therefore, image intensities are converted into Hounsfield Units (HU), which represent a standardized scale commonly used in medical imaging.

The primary objectives of the preprocessing stage are:

- Standardization of image intensity values,
- Preservation of slice ordering and spatial consistency,
- Construction of a stable and reliable representation suitable for model input.

This stage plays a critical role in the overall architecture, as the quality and consistency of preprocessing directly influence the reliability of the learned features.

6.3 Annotation-Guided Data Representation and 2.5D Input Construction

Pre-processed images are aligned with expert-provided annotation data to construct supervised learning samples. The annotations include information regarding the spatial location of lung nodules and their malignancy assessments.

Instead of employing a fully volumetric (3D) learning approach, this study adopts a 2.5D representation strategy. For each annotated nodule:

- A central slice is identified,
- Adjacent slices above and below the central slice are selected,
- The selected slices are stacked along the channel dimension to form a single input sample.

This approach enables the model to leverage limited volumetric context while maintaining

computational efficiency. Compared to purely 2D methods, the 2.5D strategy captures richer contextual information, while remaining more efficient than full 3D models.

6.4 Modeling and Learning Layer

The constructed 2.5D inputs are processed by a convolutional neural network–based classification model. The primary task of the model is to estimate the probability that a given input corresponds to a benign or malignant lung nodule.

The modeling layer encompasses both feature extraction and decision-making processes. It is treated as an independent architectural component, allowing future replacement or extension of the backbone architecture without altering the surrounding system pipeline.

6.5 Training Process and Experiment Management

The training process is managed through a structured control mechanism decoupled from the data processing pipeline. During training:

- Samples are fed to the model in a controlled manner,
- Loss values are computed and optimized through backpropagation,
- Performance metrics are monitored throughout the training process.

Separating experimental configuration from implementation logic allows systematic comparison of different training setups. This approach improves reproducibility and enhances the reliability of experimental findings.

6.6 BERTurk-Assisted Turkish NLP Layer

This section describes the BERTurk-assisted Turkish natural language processing (NLP) component integrated into the LungXai framework and its role within the clinical reporting pipeline. Unlike the image-based deep learning modules, this component focuses on extracting supportive clinical information from Turkish medical text and integrating these findings into the rule-based reporting workflow.

6.6.1 Motivation for Turkish NLP Integration

Clinical radiology workflows frequently contain free-text information such as physician notes,

patient history, smoking status, and morphology-related descriptions. These textual findings may provide clinically relevant contextual information that cannot always be directly inferred from imaging data alone.

To support Turkish clinical text analysis, the LungXai system incorporates a BERTurk-assisted NLP layer capable of processing Turkish medical expressions and extracting supportive semantic signals from textual inputs.

The primary objectives of this integration are:

- to support automated Turkish clinical reporting,
- to extract clinically relevant auxiliary information,
- to improve report completeness,
- and to maintain compatibility with rule-based recommendation generation.

6.6.2 Role of BERTurk Within the System

The BERTurk component functions as a supportive auxiliary analysis module rather than a primary diagnostic decision-maker.

The NLP layer is used for extracting information such as:

- smoking history,
- malignancy-related risk expressions,
- morphology-related descriptions,
- and other clinically relevant textual findings.

The extracted outputs are converted into structured supportive fields and integrated into the reporting workflow.

Importantly, the NLP outputs do not directly determine the final clinical recommendation. Instead, the final follow-up recommendation remains controlled by the deterministic Fleischner-based rule engine in order to preserve:

- interpretability,
- consistency,
- and clinical traceability.

6.6.3 Text Processing Workflow

The Turkish NLP pipeline follows a structured processing workflow:

1. Turkish clinical text or physician notes are received as input.
2. The text is tokenized and processed using the BERTurk language model.

3. Risk-related and morphology-related expressions are identified.
4. Extracted findings are converted into structured intermediate representations.
5. The structured outputs are transferred to the reporting layer as supportive metadata.

This workflow allows the system to combine image-based AI outputs with clinically relevant textual context.

6.6.4 Advantages of the BERTurk-Assisted Approach

The use of BERTurk provides several advantages for Turkish clinical NLP integration:

- improved compatibility with Turkish medical terminology,
- contextual understanding of Turkish free-text expressions,
- reduced dependency on manually defined keyword matching,
- and extensibility for future clinical NLP studies.

In addition, the separation between NLP outputs and deterministic clinical recommendations improves system safety by preventing unsupported NLP predictions from directly controlling medical follow-up decisions.

6.6.5 Current Limitations

At the current stage, the BERTurk integration is used primarily for supportive semantic extraction and auxiliary report enrichment.

The NLP module:

- does not independently generate final diagnoses,
- does not replace physician interpretation,
- and does not directly override the rule-based reporting engine.

Furthermore, some advanced NLP capabilities such as:

- temporal reasoning,
- uncertainty modeling,
- and large-scale fine-tuned clinical language understanding

remain outside the scope of the current implementation and are considered future development directions.

6.7 Database Schema

The Lung Nodule Detection system utilizes a relational database structure consisting of seven main tables designed to support clinical workflows, AI-based analysis, report generation, and

system-level auditing. The overall structure of the database and the relationships between its core entities are illustrated in Figure 6.7.1.

Tables

Users

The users table stores system users. Each user has a unique identifier, username, password, first name, last name, and email address. Users are assigned either Admin or Doctor roles. Professional information such as specialization, department, hospital, and license number is also maintained to support role-based access control.

Patients

The patients table contains patient records. Each patient has a unique patient_id, name, age, and gender. When a patient is registered in the system, a creation timestamp is automatically assigned.

Studies

The studies table represents CT scan studies. Each study is linked to a patient through the patient_id field, which references the patients table. Study date, description, clinical notes, and detected nodule count are stored. The study status (pending, analyzing, completed, reviewed) is maintained in the status field. When a doctor reviews a study, the reviewed_by field references the reviewing physician's identifier in the users table.

Dicom_Files

The dicom_files table stores uploaded medical imaging files. Each file belongs to a study and is linked through the study_id field to the studies table. File path, file name, and DICOM instance number are stored.

Nodules

The nodules table contains lung nodules detected by the AI model. Each nodule belongs to a study and stores anatomical location, size (in millimeters), malignancy risk level (Low, Medium, High), three-dimensional coordinates, and CT slice index. The AI confidence score is stored in the probability field. Physician assessment and notes are recorded in the doctor_assessment and notes fields. The include_in_report flag determines whether the nodule is included in the final report.

Reports

The reports table stores generated reports. Each report is linked to a study and a patient through the study_id and patient_id fields. The user who generated the report is referenced via the generated_by_id field. Total nodule count and included nodule count are stored separately. Report content is maintained in structured JSON format in the report_data field.

Activity_Logs

The activity_logs table records all user actions performed within the system. Each record references the initiating user via the user_id field and stores action type (e.g., LOGIN, UPLOAD, ANALYZE, REPORT) along with additional details for auditing purposes.

Relationships and Data Integrity

The database schema contains eight foreign key relationships. The studies table is linked to the patients table through patient_id and to the users table through reviewed_by. The dicom_files and nodules tables reference the studies table via study_id. The reports table establishes three relationships, linking to studies, patients, and users through study_id, patient_id, and generated_by_id, respectively. The activity_logs table references the users table through user_id.

Cascade delete rules are implemented to preserve data consistency. When a patient is deleted, all associated studies, DICOM files, nodules, and reports are automatically removed. When a user is deleted, related references are set to NULL to prevent unintended data loss while preserving historical records.

epoch, while the corresponding validation accuracy was approximately 0.489. These values indicate that the baseline model is able to discriminate benign and malignant nodules above random chance, despite limited accuracy at the default decision threshold.

7.2 Segmentation Results

The segmentation stage served as the primary candidate localization component of the proposed system. Its role was to identify suspicious regions in lung CT scans and provide structured candidate outputs that could later be reviewed in the interface and further evaluated by the classification module. Rather than directly producing a final diagnosis, this stage was designed to answer the question of where a possible nodule candidate might be located within the CT volume.

During development, the initial inference strategy based on fixed-size sliding-window scanning over full CT slices did not produce sufficiently reliable results. In several cases, even visually noticeable nodules were not ranked correctly, while irrelevant structures occasionally produced stronger responses. After examining the training characteristics of the model, it became clear that the checkpoint had been trained mainly on ROI-style nodule-centered crops rather than arbitrary full-slice windows. Because of this mismatch between training and inference distributions, the original segmentation outputs were less stable than expected.

To address this issue, the segmentation inference pipeline was redesigned to better reflect the model's training conditions. A multi-scale ROI-style candidate scanning strategy was introduced, allowing the system to evaluate multiple crop sizes instead of relying on a single fixed window. In addition, candidate extraction was improved by connected-component analysis, more compact candidate selection, overlap handling, and temporal consistency across adjacent slices. These adjustments significantly improved the localization quality of the segmentation stage.

After these improvements, the segmentation module became much more reliable in identifying visible suspicious regions. The system was able to localize candidate nodules more consistently and provide outputs that aligned better with the expected anatomical regions. In particular, the refined post-processing pipeline reduced the influence of noisy isolated activations and helped prioritize candidates that were more spatially compact and temporally consistent across nearby slices. A representative qualitative example of the final segmentation output is shown in Figure 7.2.1, where the suspicious lung region is successfully localized and preserved throughout preprocessing and overlay generation.



Figure 7.2.1 Representative segmentation result showing the raw CT slice, CT with segmentation overlay, candidate-centered preprocessing crop, predicted mask, and overlay visualization. The figure demonstrates that the segmentation model successfully localizes the suspicious nodule region while preserving spatial alignment.

The final output of the segmentation stage was not treated as a direct medical conclusion, but as a set of structured AI-generated candidates. Each candidate included slice information, a bounding box, a candidate center, segmentation confidence, and mask area. This made the segmentation output usable not only for backend inference but also for practical frontend review. The results were displayed on the review page, where users could inspect all DICOM slices, navigate directly to candidate-containing slices, and visualize the segmented regions through the segmentation overlay.

An important improvement in the final system was the redesign of the segmentation overlay generation. Earlier overlay versions unintentionally contained the CT image itself, which caused visual artifacts when rendered on top of the DICOM viewer. This problem was resolved by generating transparent overlays containing only contours, masks, and bounding boxes. As a result, the final segmentation visualization highlighted only the candidate region without degrading the underlying CT image quality. This significantly improved the clarity and usability of the review interface.

Another important result of the segmentation stage was the conversion of candidate size information from pixel units into physically meaningful millimeter values. Since raw pixel counts are not directly comparable across DICOM studies, the system was updated to use DICOM pixel spacing metadata. In the final version, candidate size was represented using equivalent diameter calculated from the segmented mask area. This provided a more interpretable physical size estimate and made the reported segmentation result more suitable for clinical-style review and reporting.

In addition to qualitative observations, the training behavior of the SegResNet-based segmentation model also indicated stable optimization. As illustrated in Figure 7.2.2, both training and validation loss decreased over the course of training, while Dice scores increased steadily. This trend suggests that the model was able to learn meaningful localization patterns from the training data and improve its segmentation performance consistently. The validation Dice curve, in particular, showed that the model maintained a stable generalization trend rather than learning only the training samples.

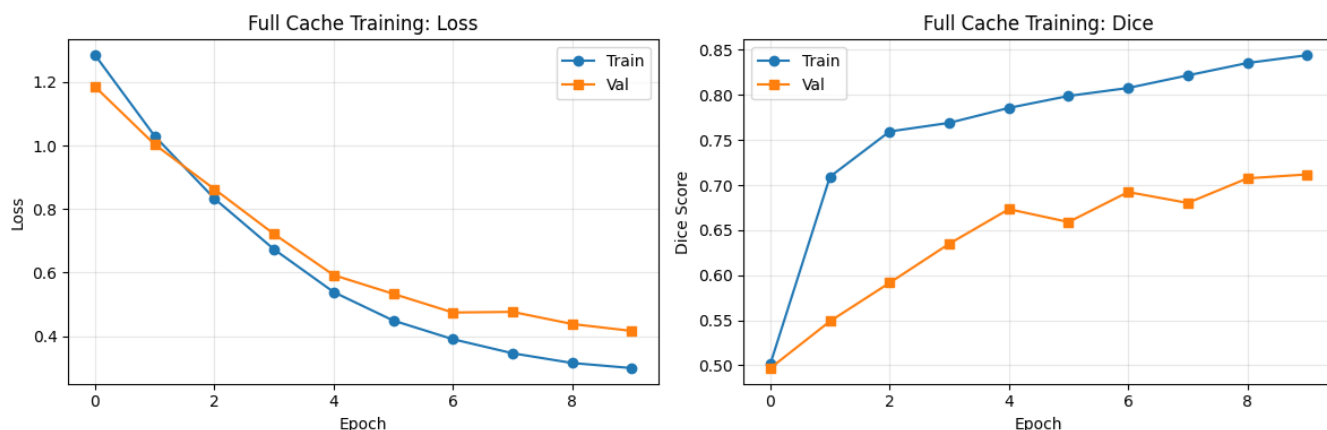


Figure 7.2.2 Training and validation curves of the SegResNet-based segmentation model. The decreasing loss values and increasing Dice scores indicate stable optimization and improved segmentation performance throughout training.

To further illustrate the contextual nature of the segmentation process, Figure 7.2.3 presents an example of the candidate-centered crop and the corresponding neighboring slices used in the 2.5D segmentation setup. This visualization highlights that the model does not rely solely on a single slice, but instead benefits from adjacent slice information when localizing suspicious regions. Such context-aware processing improves robustness, especially for nodules that extend across multiple slices or appear with varying visibility in neighboring sections.

Overall, the segmentation module successfully fulfilled its intended role as the first decision layer of the system. It localized suspicious candidate regions, provided structured outputs for subsequent classification, enabled visual review in the frontend, and contributed physically meaningful measurements for reporting. Although it should not be interpreted as a standalone diagnostic tool, it formed the essential backbone of the full lung nodule analysis pipeline

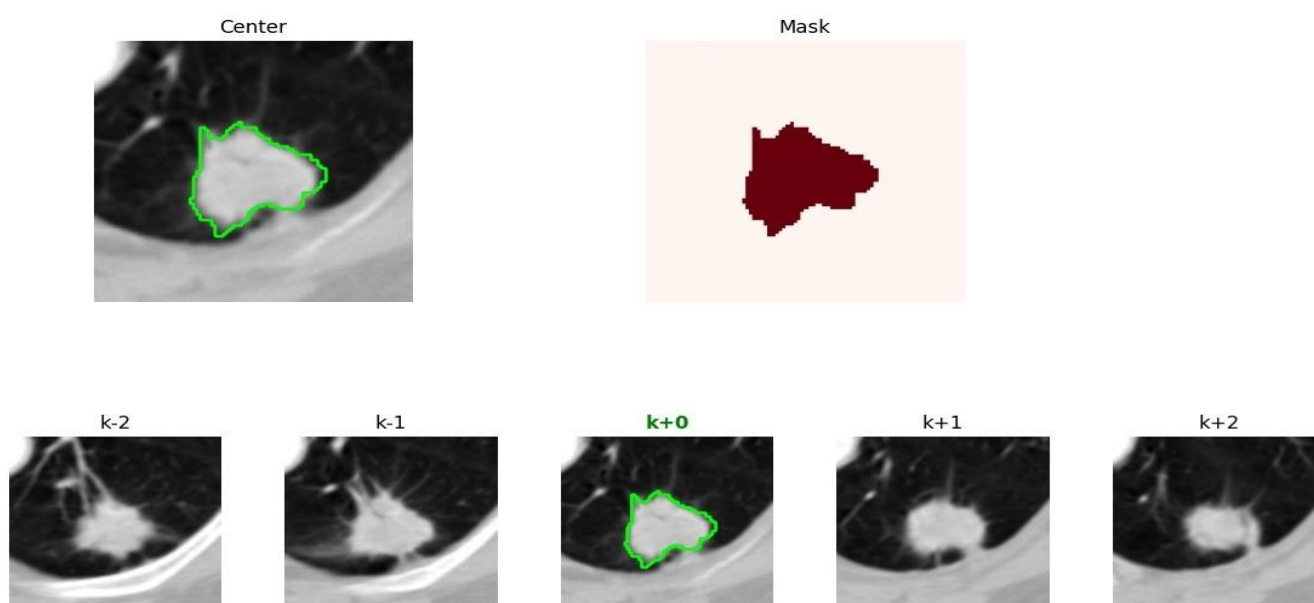


Figure 7.2.3 Example of candidate-centered preprocessing and neighboring slice context used in the 2.5D segmentation pipeline. The figure illustrates how the segmentation model benefits from adjacent slice information while preserving the target mask region.

7.3 Classification Results

The classification phase of the LungXai framework is built upon a highly optimized teacher-student knowledge distillation architecture. This approach was specifically designed to overcome the computational bottlenecks typically associated with volumetric medical image analysis, bridging the gap between high diagnostic fidelity and practical deployment constraints.

Initially, a robust 3D baseline was established using the SegResNet architecture. For this stage, we utilized the LUNA16 dataset, a rigorously curated subset of the LIDC-IDRI database, to train and evaluate the model. The 3D SegResNet successfully captured the complex morphological variations of pulmonary nodules, achieving a validation accuracy of 90%. By processing full 3D patches, the network developed a comprehensive understanding of the volumetric context and the surrounding pulmonary microenvironment.

However, deploying such a high-capacity 3D network introduces significant memory and computational overhead, limiting its real-world clinical scalability. To address this, we implemented a knowledge distillation strategy. We transferred the learned representations from the computationally heavy 3D teacher to a much lighter 2.5D student model. This student model processes tri-planar representations, specifically axial, coronal, and sagittal slices, to approximate the full volumetric data efficiently.

This structural adaptation represents a critical paradigm shift in our methodology. By guiding the 2.5D student to mimic the high-level feature maps of the 3D teacher, we preserved essential 3D contextual information while drastically reducing the computational footprint. The learning objective was driven by a combined loss function that balances standard classification with the distillation process, formalized as Figure 7.3.1:

$$L_{total} = \alpha L_{CE} + (1 - \alpha) L_{KD}$$

Figure 7.3.1 The loss function that balances standard classification with the distillation process

As a result, the 2.5D model operates with a fraction of the VRAM and inference latency required by its 3D counterpart. This confirms that multi-planar spatial awareness can be effectively distilled without the need for fully volumetric convolutions, maximizing diagnostic yield while minimizing hardware requirements.

Despite the success of this distillation approach, a rigorous assessment requires acknowledging certain architectural trade-offs. First, while the 3D teacher achieved 90% accuracy, relying solely on accuracy in imbalanced medical datasets can be misleading. Therefore, emphasizing the ROC-AUC metric remains crucial for evaluating the true discriminative power of the student model. Furthermore, projecting a rich 3D feature space into a 2.5D tri-planar student inevitably results in a minor loss of fine-grained spatial continuity, which may marginally affect the evaluation of highly irregular subsolid nodules. Finally, the performance of the 2.5D student is inherently bound by the teacher's initial feature extraction capabilities; any localized feature collapse during the SegResNet stage will implicitly propagate to the student during distillation.

7.4 Grad-CAM Results

We conducted numerous trials and tests to integrate Grad-CAM, also known as XAI (Explainable Artificial Intelligence), into our system. First and foremost, to fully grasp the mechanics of Grad-CAM, we attempted to make it focus solely on an inner circle by simply embedding a smaller circular shape inside a rudimentary outer one. In relation to this, some initial preliminary visual representations are presented below:

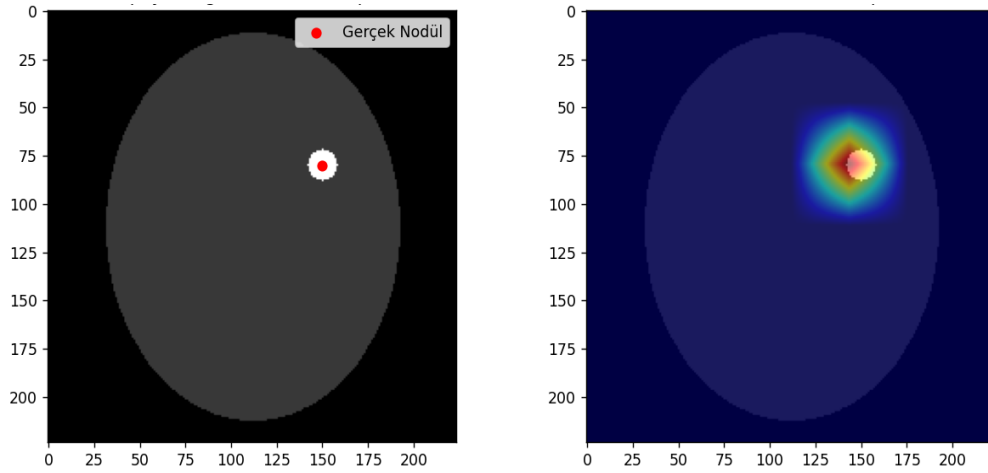


Figure 7.4.1 The initial test result that successfully identified the correct focus point based on color contrast.

Consequently, this process enabled us to comprehend the fundamental functions and algorithms required to effectively utilize Grad-CAM.

Subsequently, when we initiated testing with actual DICOM-formatted masked .npy and .npz datasets alongside our trained models using the same algorithms and functions, Grad-CAM began producing erratic results. This occurred because those masks contained not just two distinct areas, but also extraneous micro-regions; moreover, Grad-CAM was scanning the entire space, including regions with the lowest activation. To address this, impose boundaries, and further refine the visualization, we applied thresholding to the heatmap, which severely backfired. Instead of highlighting the nodule, it mapped the non-nodular regions. Upon investigating this anomaly, we discovered it was experiencing a reverse activation paradox, technically referred to as an *Inversion Attention* error. Consequently, to aid our troubleshooting, we implemented accuracy evaluation functions and restructured certain preprocessing functions to resolve emerging computational numerical discrepancies and model score mismatches. Furthermore, we realized that the accuracy had initially been miscalculated. After rectifying all these discrepancies, the correct localization mapping was ultimately achieved. In connection with these findings, several illustrative test images are presented below:

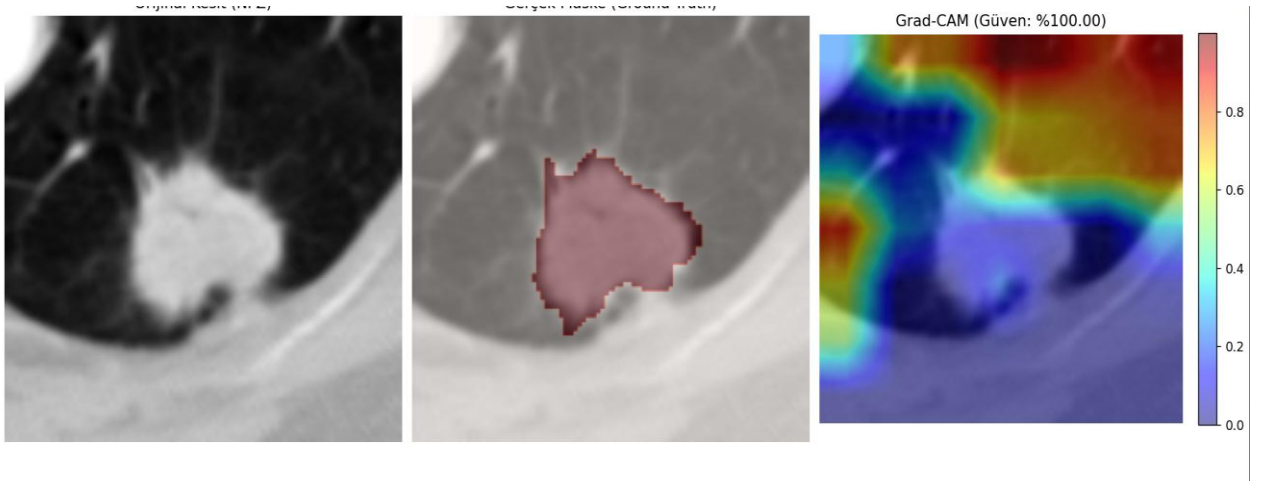


Figure 7.4.2 The initial test result demonstrating the first instance of discrepancy and model confusion.

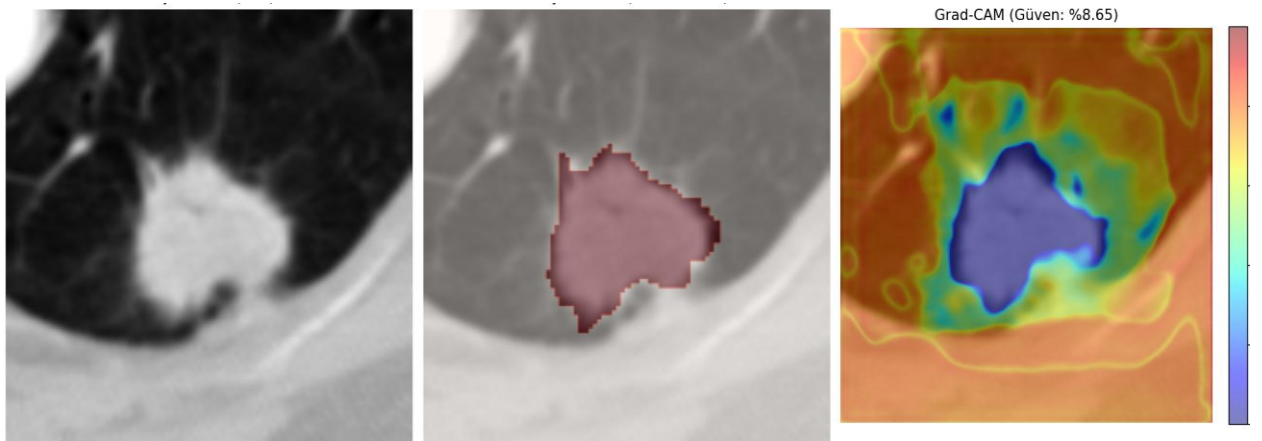


Figure 7.4.3 Test result demonstrating the Inverse Attention Paradox.



Figure 7.4.4 The test result obtained following post-adjustment refinements.

Ultimately, the mask coordinates derived from the .npy files were able to map the nodule exceptionally well within specific regions. The next objective was to achieve this localization dynamically while scrolling through the sequential DICOM slices, which necessitated pinpointing the exact correspondence between the masks and their respective slices. However, a significant bottleneck arose during the integration of the masks with the cross-sectional slice images; despite utilizing an accurate model and proper preprocessing, the system began to yield erratic outputs. Seeking a workaround, we investigated methods to re-align the mask files back into the DICOM framework. Once this integration was successfully achieved and validated through a correct test run, we deployed the final results directly onto the user interface (UI), thereby successfully concluding our Grad-CAM implementation. In connection with these findings, several illustrative test images are presented below:

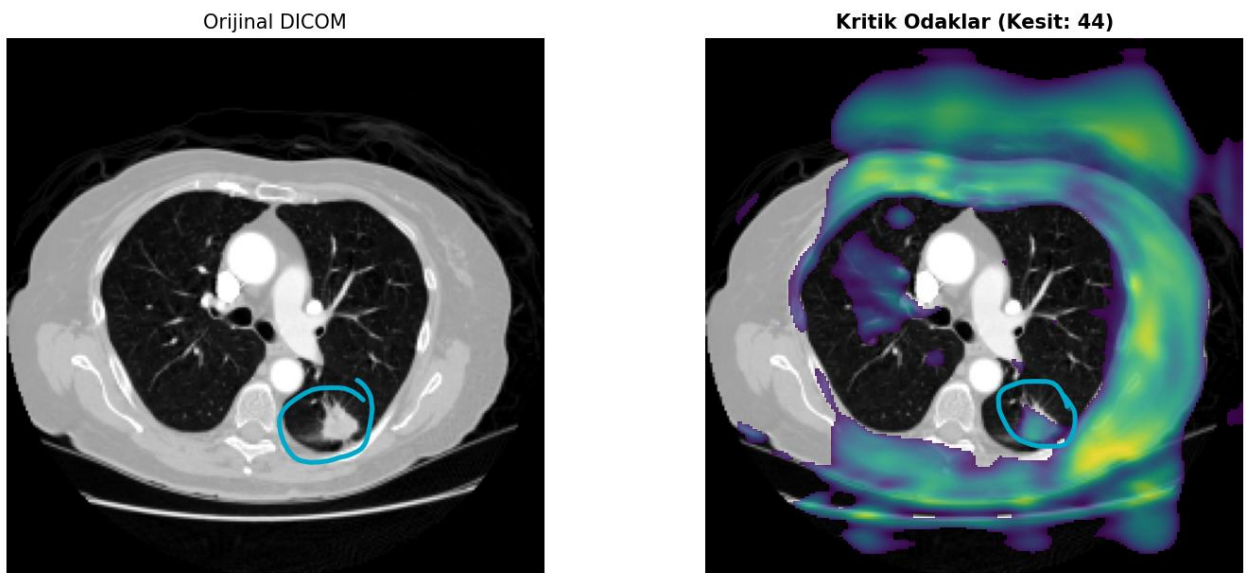


Figure 7.4.5 An erroneous test result showing over-scanning of extraneous regions due to integration issues.

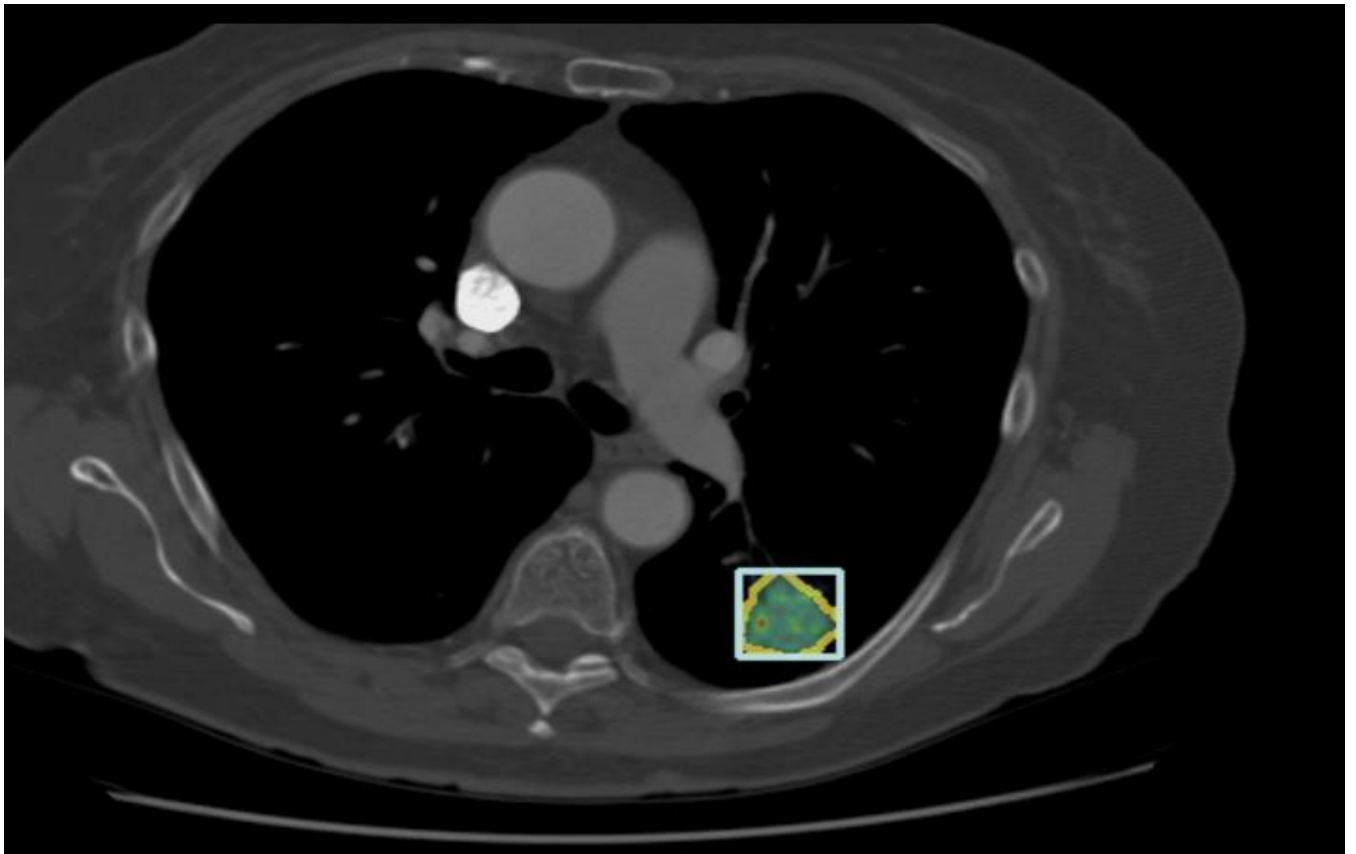


Figure 7.4.6 The UI rendering of the correct result following the resolution of the integration issue.

In conclusion, throughout the journey from the initial baseline tests to the final evaluations—even though we encountered severe errors or temporarily overlooked specific parameters at certain stages—we progressed iteratively by identifying the correct outcomes and constantly mastering, refining, and adapting various algorithms and methodologies. Ultimately, this systematic approach enabled us to successfully implement Grad-CAM, thereby achieving Explainable Artificial Intelligence (XAI) within our framework.

7.5 NLP Results

This section describes the BERTurk-assisted Turkish natural language processing (NLP) module integrated into the LungXai framework and its role within the clinical reporting workflow. Unlike the image-based deep learning components, this module focuses on extracting supportive clinical information from Turkish medical text and incorporating these findings into the automated reporting pipeline.

In clinical radiology workflows, not only imaging data but also free-text clinical information such as smoking history, physician notes, malignancy-related expressions, and morphology-related findings play an important role in decision-making. Therefore, an auxiliary NLP layer capable of understanding Turkish clinical expressions was integrated into the system. For this purpose, the BERTurk language model was utilized.

The BERTurk module is responsible for extracting supportive information such as:

- smoking history,
- malignancy-related risk expressions,
- morphology-related descriptions,
- and other clinically relevant textual findings.

The extracted outputs are not directly used for final diagnosis generation. Instead, these findings are converted into structured auxiliary fields and transferred to the rule-based reporting system as supportive metadata.

Within the system, the final clinical recommendation is generated entirely by the deterministic Fleischner-based rule engine. Therefore, the NLP module functions as a supportive component that enriches report content rather than directly making clinical decisions. This design improves system interpretability, traceability, and clinical safety.

The integration of BERTurk provides several advantages:

- improved compatibility with Turkish clinical terminology,
- contextual understanding of Turkish free-text expressions,
- reduced dependency on keyword-based approaches,
- and a modular infrastructure suitable for future clinical NLP studies.

PULMONER NODÜL RAPORU

Bilgilendirme ve izlem amaçlı ön değerlendirme özeti

HASTA VE İNCELEME BİLGİLERİ

Rapor No:	RPT-1779368805298	Oluşturma Tarihi:	21.05.2026 19:06
Oluşturan:	Demo Doctor	Nodül Paterni:	Tek nodül
Hasta Adı:	ebrar özmen	Hasta ID:	LIDC-IDRI-0263
Çalışma Tarihi:	2026-05-19	İnceleme No:	STD-1779179662848
Yaş / Cinsiyet:	62 / F		

İNCELEME ÖZETİ

2026-05-19 tarihli toraks BT incelemesi değerlendirilmiştir. Klinik bilgi olarak Sigara geçmişi var..

BULGULAR

Sağ orta lob (RML) yerleşimli, yaklaşık 22.4 mm çapında tek pulmoner nodül izlenmiştir.

DEĞERLENDİRME

Nodül boyutu, sayısı ve mevcut klinik veriler birlikte değerlendirildiğinde yakın klinik/radyolojik takip gerektirebilecek bir görünüm mevcuttur. Hasta öyküsünde sigara öyküsü, önceki kanser / onkolojik öykü, eşlik eden akciğer hastalığı / hava yolu obstrüksiyonu, cinsiyet ve yaş ilişkili risk bulunması nedeniyle izlem gereksinimi artmaktadır. Klinik notlarda sigara öyküsü, önceki kanser / onkolojik öykü, eşlik eden akciğer hastalığı / hava yolu obstrüksiyonu, cinsiyet ve yaş ilişkili risk saptandı. Bu nedenle yakın klinik değerlendirme ve ileri inceleme ihtiyacı doğabilir.

NODÜL AYRINTILARI

Nodül 38

Konum:	Sağ orta lob (RML)
Boyut:	22.4 mm
Risk Değerlendirmesi:	Yüksek risk
Ortalama Risk Oranı:	0.890
Hekim Değerlendirmesi:	malignant
Not:	KOAH rahatsızlığı var.

İZLEM VE SONUÇ

8 mm üstü tek nodül için yaklaşık 3 ay içinde kontrol BT ve gerekli klinik durumda PET/BT ya da ileri tanısal değerlendirme düşünülmelidir.

Bu belge mevcut görüntüleme ve kayıtlı klinik bilgiler kullanılarak hazırlanmıştır. Nihai değerlendirme ve tedavi planı ilgili uzman hekim tarafından yapılmalıdır.

Figure 7.5.1 illustrates the workflow of the BERTurk-assisted NLP module for extracting supportive risk-related information from Turkish clinical text.

PULMONER NODÜL RAPORU	
Bilgilendirme ve izlem amaçlı ön değerlendirme özeti	
HASTA VE İNCELEME BİLGİLERİ	
Rapor No:	RPT-1779356029854
Oluşturma Tarihi:	21.05.2026 15:33
Oluşturan:	Demo Qeapax
Nodül Patemi:	Çoklu nodül
Hasta Adı:	Qeapax
Hasta ID:	LIDC-IDRI-0005
Çalışma Tarihi:	2026-05-19
İnceleme No:	STD-1779177574969
Yaş / Cinsiyet:	30 / M
İNCELEME ÖZETİ	
2026-05-19 tarihli toraks BT incelemesi değerlendirilmiştir. Klinik bilgi olarak Hastanın sigara geçmişi var .	
BULGULAR	
Her iki akciğer değerlendirilmesinde birden fazla Qeapax nodül izlenmiştir. En büyük nodül Sol alt lob (LLL) yerleşimli olup yaklaşık 26.4 mm çapındadır.	
DEĞERLENDİRME	
Nodül boyutu, sayısı ve mevcut klinik veriler birlikte değerlendirildiğinde yakın klinik/radyolojik takip gerektirebilecek bir görünüm mevcuttur. Hasta öyküsünde sigara öyküsü, öncelikli kanser / onkolojik öykü, eşlik eden akciğer hastalığı / hava yolu obstrüksiyonu ilişkili risk bulunması nedeniyle izlem gereksinimi artmaktadır. Klinik notlarda sigara öyküsü, öncelikli kanser / onkolojik öykü, eşlik eden akciğer hastalığı / hava yolu obstrüksiyonu, cinsiyet ve yaş ilişkili risk saptandı. Bu nedenle yakın klinik değerlendirme ve ileri inceleme ihtiyacı doğabilir.	
NODÜL AYRINTILARI	
Nodül 36	
Konum:	Sol alt lob (LLL)
Boyut:	23.2 mm
Risk Değerlendirmesi:	Orta risk
Ortalama Risk Oranı:	0.880
Hekim Değerlendirmesi:	Qeapax
Not:	KOAH rahatsızlığı
Sayfa 1 / 2	

Nodül 37	
Konum:	Sol alt lob (LLL)
Boyut:	26.4 mm
Risk Değerlendirmesi:	Orta risk
Ortalama Risk Oranı:	0.850
Hekim Değerlendirmesi:	Qeapax
Nodül 35	
Konum:	Sol alt lob (LLL)
Boyut:	19.5 mm
Risk Değerlendirmesi:	Orta risk
Ortalama Risk Oranı:	0.880
İZLEM VE SONUÇ	
8 mm üstü ya da baskın büyük nodül varlığında 3-6 ay içinde kontrol BT ve gerekli görülürse ileri değerlendirme planlanması uygundur.	
Bu belge mevcut görüntüleme ve kayıtlı klinik bilgiler kullanılarak hazırlanmıştır. Nihai değerlendirme ve tedavi planı ilgili uzman hekim tarafından yapılmalıdır.	
Sayfa 2 / 2	

Figure 7.5.2 presents an example Turkish clinical report generated by the system.

However, at the current stage, BERTurk is used only as a supportive NLP component. The system:

- does not independently diagnose nodules,
- does not replace physician interpretation,
- and does not directly override image-based clinical decisions.

Furthermore, advanced capabilities such as temporal clinical reasoning, uncertainty modeling, and large-scale fine-tuned clinical language understanding remain outside the scope of the current study and are considered future development directions.

7.6 Observer Study

In order to clinically evaluate the outputs of the proposed deep learning model, an observer study was conducted with a radiologist. The detected regions produced by the model were reviewed together, and the radiologist provided professional feedback regarding whether the highlighted areas could be considered true pulmonary nodules or false positive findings. The comments below summarize the

radiologist's evaluations and interpretations for each patient case.

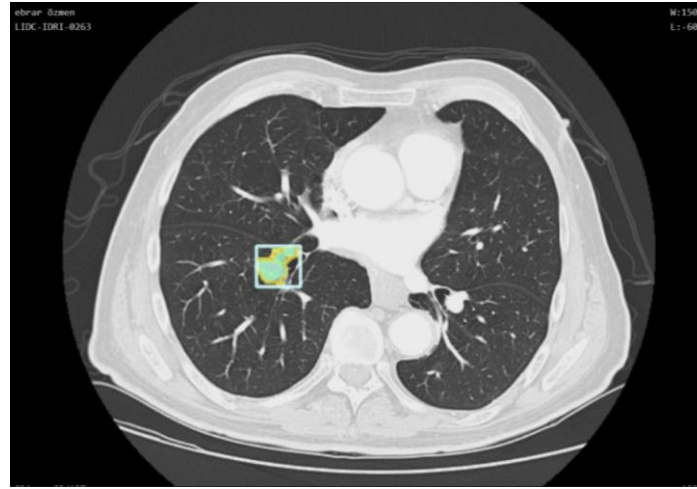


Figure 7.6.1 – The View of Observer on Patient 0263

The highlighted region in the upper left lung was evaluated by the observer as a possible vascular continuation rather than a true pulmonary nodule. Sequential slice inspection was recommended to confirm the finding.

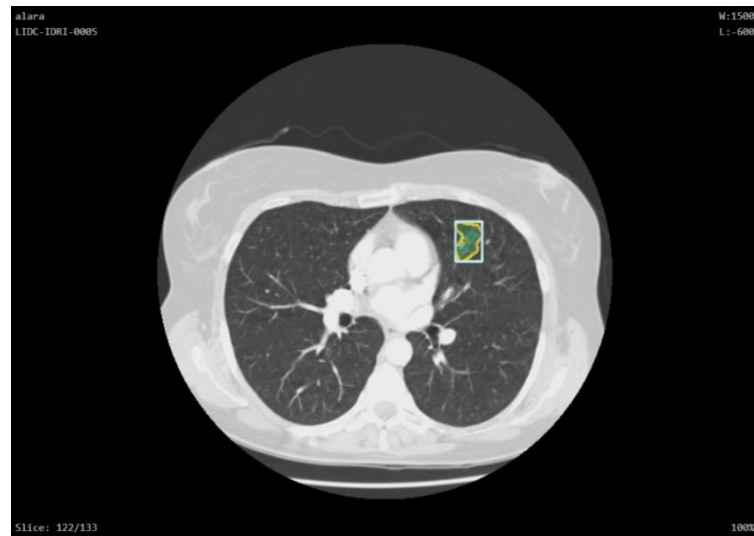


Figure 7.6.2 – The View of Observer on Patient 0005

No clear pulmonary nodule was identified by the observer in the marked region. This case was considered a potential false positive prediction of the model.

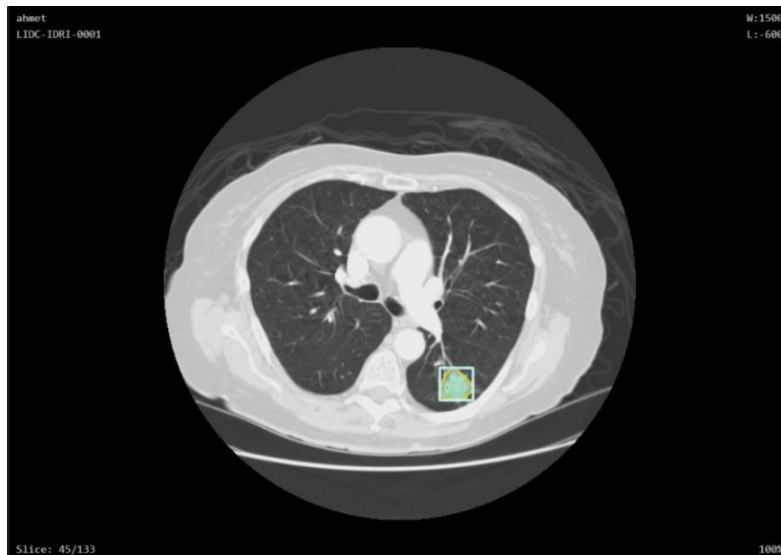


Figure 7.6.3 – The View of Observer on Patient 0001

The observer agreed with the model prediction for this patient. The highlighted area was considered compatible with a pulmonary nodule appearance.

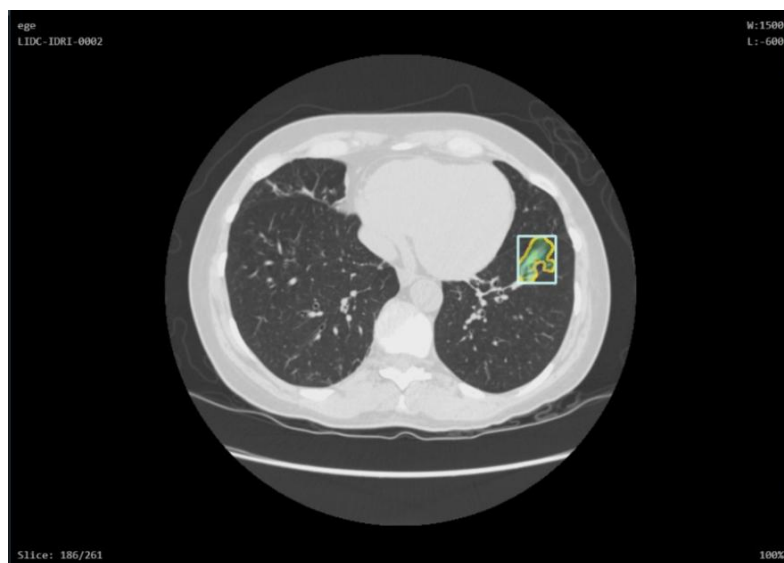


Figure 7.6.4 – The View of Observer on Patient 0002

The observer stated that the detected region should not be directly classified as a pulmonary nodule. The finding was interpreted as a non-nodular structure.

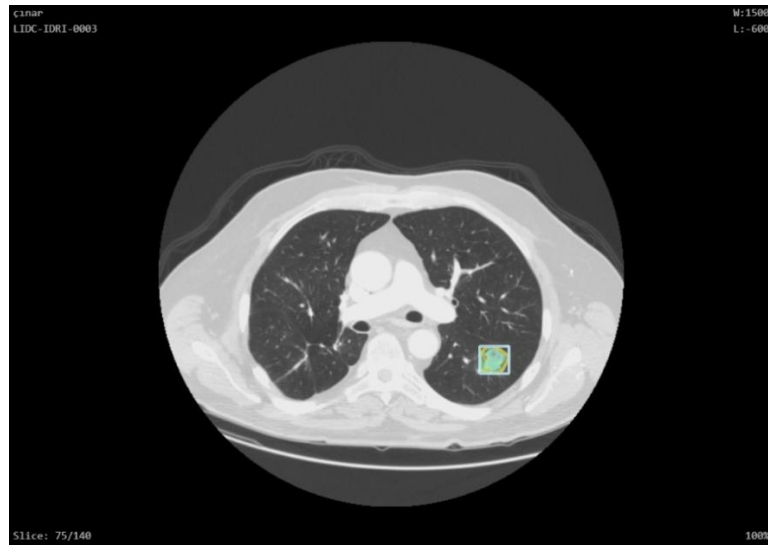


Figure 7.6.3 – The View of Observer on Patient 0003

The detected region demonstrated ground-glass opacity characteristics. The observer suggested that the appearance could be associated with infection; however, a semisolid nodule possibility could not be completely excluded. Follow-up imaging was recommended.

8. User Interface (UI) Module

This section describes the User Interface (UI) module developed as part of the LungXai system. The UI serves as an interaction layer between the trained deep learning models and end users, enabling model outputs to be accessed, interpreted, and reviewed in a structured and user-friendly manner. Rather than targeting direct clinical deployment, the interface is designed as a demonstration-oriented and research-focused visualization tool.

8.1 Purpose and Design Goals

The primary objective of the UI module is to demonstrate how the proposed lung nodule classification system can be integrated into a practical software environment. The interface is designed to:

- Provide a clear workflow for reviewing imaging-based analysis results
- Present model predictions in an interpretable and accessible format
- Support inspection of multiple cases and system configurations

- Separate presentation logic from model inference logic

The design prioritizes clarity, modularity, and extensibility, allowing future integration of additional outputs such as segmentation masks or localization results.

8.2 UI Architecture Overview

The UI is implemented as a modular, component-based web application. Its architecture follows a clear separation of concerns:

- **Layout Components:**

A persistent layout structure organizes the interface into navigation and content regions. This includes a top navigation bar and a side navigation panel, enabling consistent access to system pages.

- **Page-Level Views:**

Individual pages correspond to distinct system functionalities, such as work list management, case review, system settings, and user management.

- **Service Layer:**

A dedicated communication layer handles interactions with backend services or APIs, abstracting data retrieval and submission from the presentation layer.

- **Utility Layer:**

Supporting utilities manage domain-specific operations such as DICOM-related handling and data formatting.

This layered UI architecture ensures maintainability and allows interface elements to evolve independently from backend or model-level changes.

8.3 User Workflow and Interaction Scenario

A typical user interaction with the LungXai UI follows the workflow outlined below:

1-Case Selection:

The user accesses a work list containing available imaging cases. Each entry represents a study or sample ready for review. As shown in Figure 8.3.1, users are required to authenticate

through a dedicated login interface before accessing any system functionality.

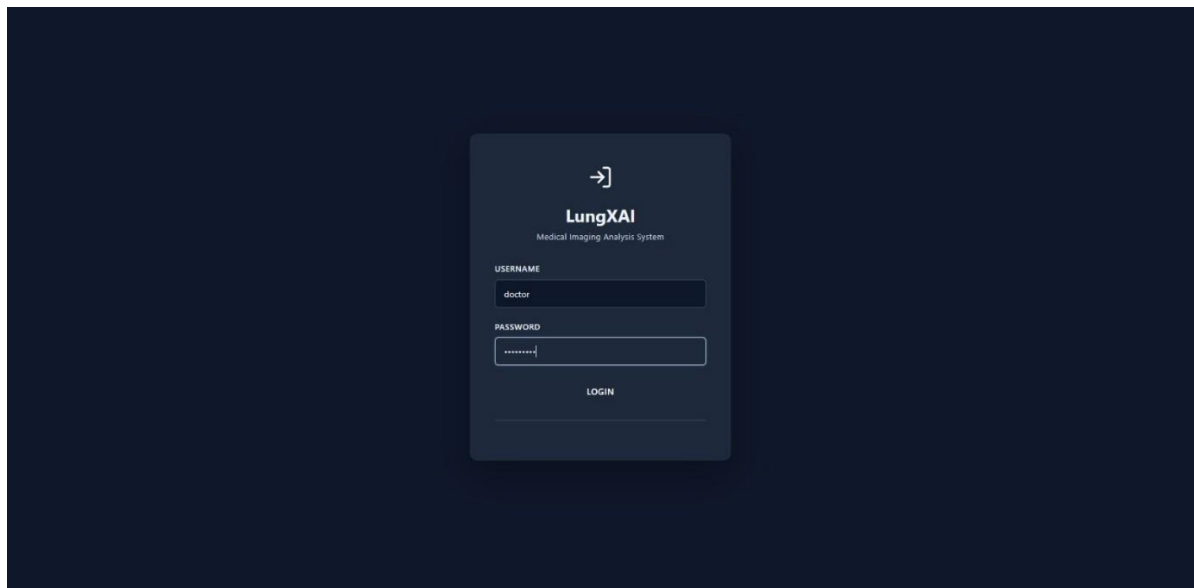


Figure 8.3.1: Login interface of the LungXai system, providing role-based authentication for authorized users.

Case Review:

Upon selecting a case, the user is navigated to a review interface where imaging data and associated metadata are displayed. After authentication, the user initiates a new study by entering patient information and uploading DICOM files, as shown in Figure 8.3.2.

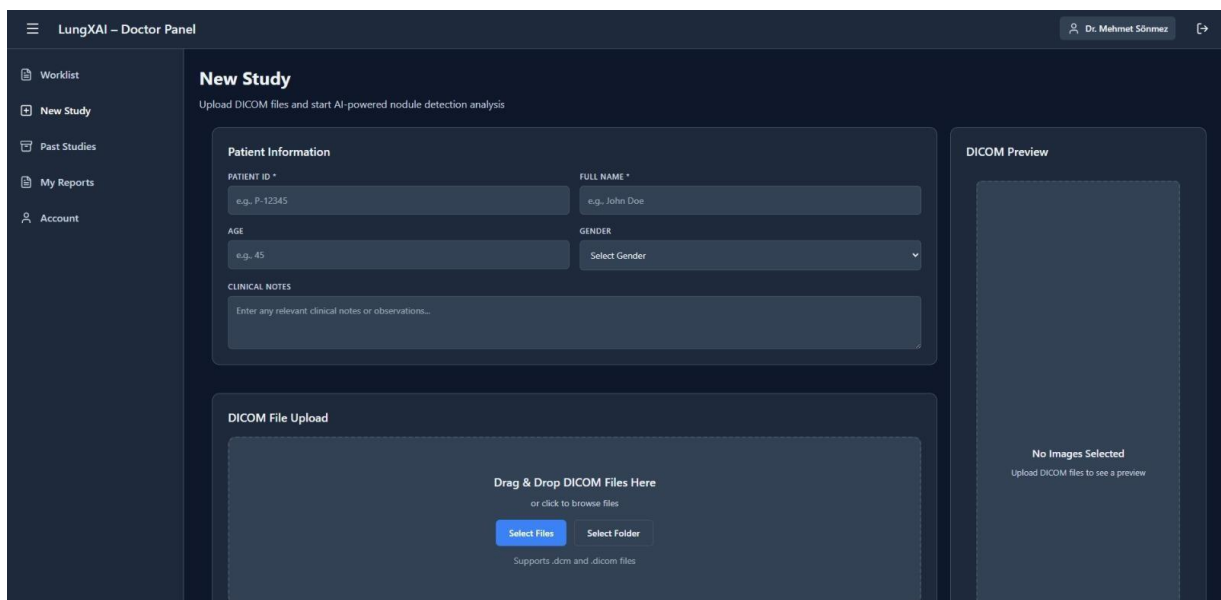


Figure 8.3.2: Interface for creating a new study, including patient information entry and DICOM file upload for AI-based lung nodule analysis.

Model Output Visualization:

The UI presents the model's classification output, including predicted class labels and confidence-related information.

System Navigation:

The user may navigate between different sections of the interface, such as system configuration panels or administrative pages, without interrupting the review workflow.

This scenario reflects a realistic usage pattern for decision support systems in research or clinical validation settings.

8.4 Navigation and Interface Components

The interface includes a consistent navigation system that allows users to move efficiently between different functional areas. Key UI components include:

- A sidebar navigation panel for accessing major system modules
- A top bar for contextual actions and system-level information
- Reusable layout elements that ensure visual consistency across pages

Such component reuse simplifies maintenance and provides a cohesive user experience.

Figure 8.5.2 shows the administrative dashboard, which supports system monitoring, user management, and navigation across non-clinical functionalities.

8.5 Presentation of Model Outputs

Model predictions are displayed in a structured and readable manner. Rather than exposing raw numerical outputs, the UI emphasizes interpretability by:

- Clearly indicating the predicted class (e.g., benign or malignant)
- Associating predictions with confidence or probability values
- Presenting results alongside the corresponding imaging context

This design ensures that outputs are not interpreted in isolation but are visually and contextually linked to the underlying CT data. As shown in Figure 8.5.1, model predictions are presented together with the corresponding CT slice, allowing clinicians to interpret AI outputs in direct anatomical context.

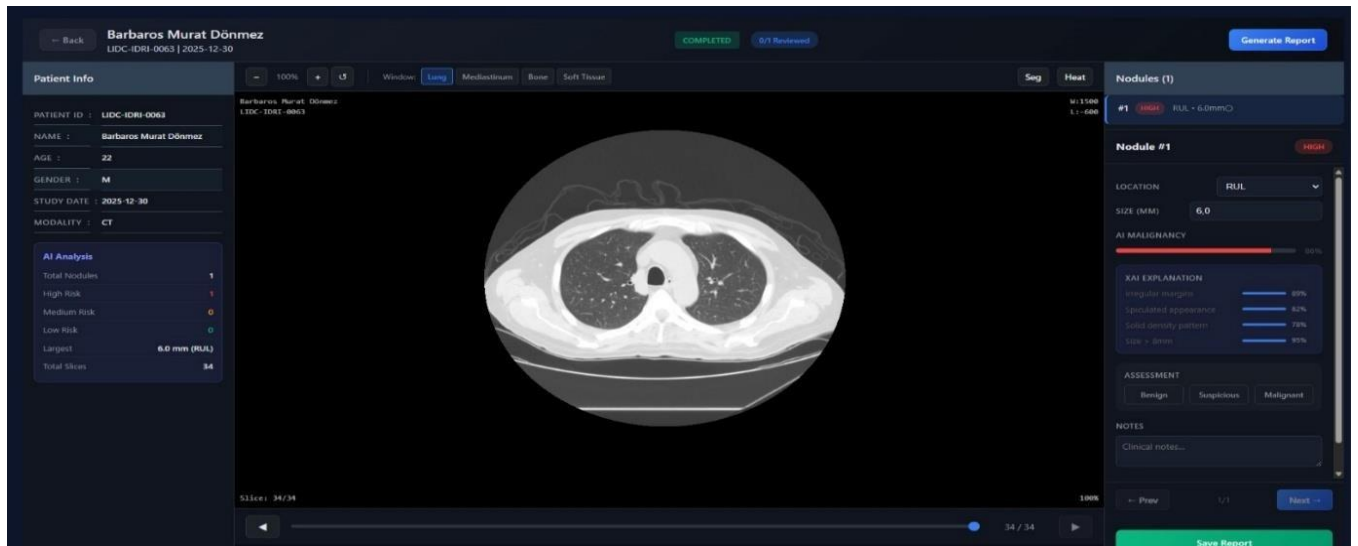


Figure 8.5.1: Visualization of model predictions displayed alongside the corresponding CT slice, including malignancy assessment and confidence indicators.

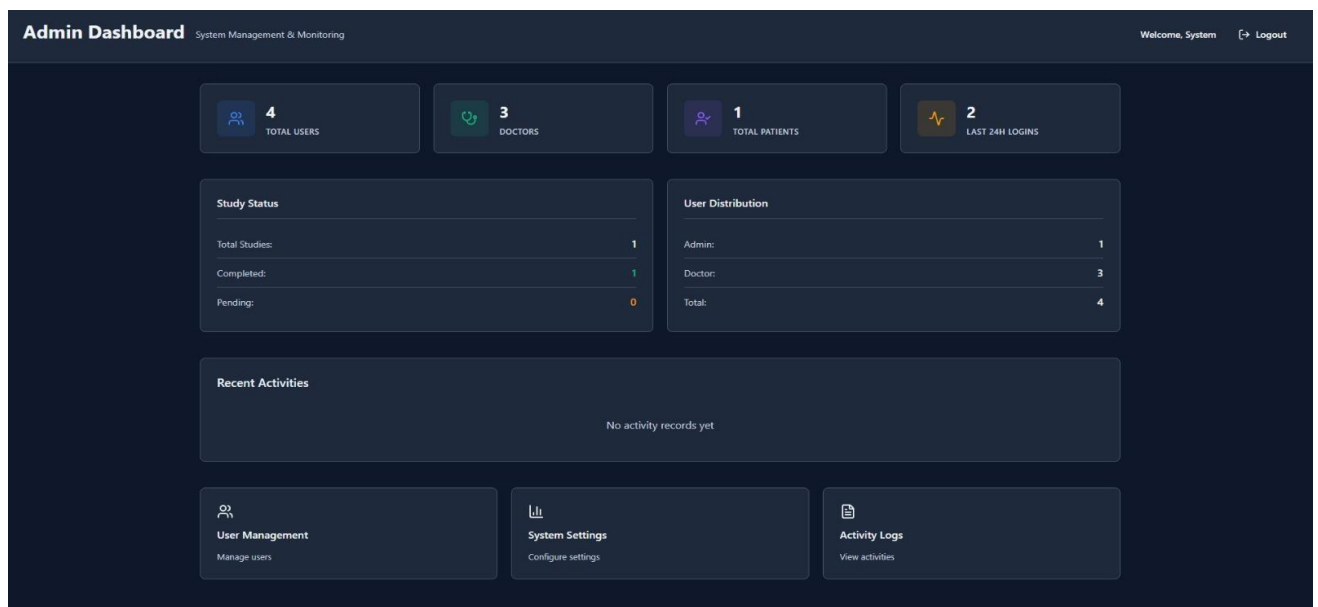


Figure 8.5.2: Administrative dashboard providing system monitoring, user management, and navigation across non-clinical functionalities.

8.6 Role of the UI within the LungXai System

Within the overall LungXai architecture, the UI module functions as a visualization and interaction layer rather than a core analytical component. Its role is to:

- Demonstrate end-to-end integration of data, models, and outputs
- Facilitate qualitative inspection of model behavior
- Support future extensions such as visualization of segmentation masks or localized predictions

By decoupling the UI from model internals, the system remains flexible and adaptable to evolving research objectives.

8.7 Limitations and Future Extensions

The current UI focuses on classification output presentation and system navigation. Planned future enhancements include:

- Visualization of segmentation results overlaid on CT images
- Display of predicted nodule locations (e.g., coordinate-based markers)
- Enhanced case comparison and filtering tools
- Integration of explainability outputs such as heatmaps

These extensions would further improve interpretability and usability, particularly for expert users.

8.8 Use Case Diagram

The use case diagram summarizes the interaction between different user roles and the main functional components of the LungXai user interface. The diagram is designed to present who can perform which actions within the system, rather than illustrating the execution order of operations.

8.8.1 User Roles (Actors)

The LungXai system defines two primary user roles:

- **Doctor**

Represents clinical users who interact with medical imaging data, initiate analysis sessions, review CT images, and interpret AI-generated malignancy predictions.

- **Admin**

Represents system administrators responsible for managing users, configuring system settings, and monitoring overall system activity and performance.

These roles are intentionally separated to ensure a clear distinction between clinical decision-making workflows and administrative system management.

8.8.2 Common Use Case: Authentication

Authentication is a mandatory prerequisite for accessing any system functionality. Both Doctor and Admin actors must successfully authenticate through the Login use case before interacting with protected features.

In the use case diagram, authentication is modeled as a shared mandatory behavior using <<include>> relationships. This modeling choice reflects role-based access control (RBAC) rather than execution order, ensuring that system functionalities are available only to authorized users.

8.8.3 Doctor Use Cases

- **Create New Study:** Allows the doctor to initiate a new analysis session by entering patient information and preparing the case for evaluation.
- **Upload DICOM Files:** Enables the uploading of medical imaging data required for AI-based lung nodule analysis.
- **Preview DICOM Files:** Allows optional visual inspection of uploaded images prior to analysis.
- **View CT Images:** Enables interactive browsing and inspection of CT slices.
- **View AI Malignancy Prediction:** Displays malignancy risk estimates, confidence indicators, and explainable AI outputs.
- **Review Case:** Provides an integrated environment for clinical assessment by combining CT image visualization and AI-generated malignancy predictions.

- **Generate Report:** Allows the doctor to generate a structured medical report summarizing imaging findings and AI-assisted assessments.
- **Save Report:** Persists the generated report for later access and review.

Mandatory clinical actions, such as viewing CT images and AI predictions, are modeled using <<include>>, while optional behaviors, such as report generation and clinical note entry, are modeled using <<extend>> relationships.

8.8.4 Admin Use Cases

Admin use cases are associated exclusively with system management and monitoring functionalities:

- **View Dashboard:** Provides a high-level overview of system usage and activity summaries.
- **Manage Users:** Allows the creation, modification, and management of user accounts and roles.
- **View System Statistics:** Enables monitoring of performance metrics and usage data.
- **View Activity Logs:** Supports auditing and traceability through access to system event logs.
- **Configure System Settings:** Enables adjustment of system-level configurations without interfering with clinical workflows.

Administrative use cases are intentionally isolated from medical decision processes to maintain system security and workflow integrity.

8.8.5 Use Case Diagram Overview

Figure 8.8.5.1 presents the use case diagram of the LungXai system, illustrating role-based interactions between clinical and administrative users. The diagram emphasizes functional access rights through the use of <<include>> and <<extend>> relationships, while workflow sequencing and user interface navigation are addressed in subsequent sections.

During baseline experimentation, it became evident that even relatively lightweight 2.5D classification pipelines introduce non-trivial computational overhead when applied to medical imaging data. CT-based workflows differ from natural image tasks due to volumetric data handling, medical preprocessing requirements, and evaluation with probability-based metrics.

Therefore, an exploratory optimization study was conducted to:

- Identify practical efficiency constraints,
- Understand which components dominate runtime,
- Guide future design decisions before integrating more complex modules such as segmentation and explainability.

This section intentionally avoids claiming final performance gains and instead focuses on engineering insights gained during development.

9.2 Observed Computational Constraints

Through empirical experimentation, the following constraints were identified:

- **Preprocessing overhead dominates early experimentation**, especially when repeatedly reconstructing samples from raw medical data.
- **GPU utilization is not always the primary bottleneck**; inefficient data preparation can limit throughput even with a lightweight backbone.
- **Memory usage scales with input dimensionality**, making future extensions (e.g., segmentation or multitask heads) sensitive to VRAM constraints.

These observations motivated targeted investigations rather than global architectural changes.

9.3 Explored Efficiency-Oriented Strategies

Several efficiency-oriented strategies were examined at both conceptual and experimental levels to better understand the computational characteristics of the proposed system.

9.3.1 Reuse of Preprocessed Data

Since many preprocessing steps are deterministic, reusing intermediate representations across epochs was explored to reduce redundant computation. This strategy is particularly relevant for medical datasets where raw input processing is significantly more expensive than model forward passes.

9.3.2 Training Configuration Sensitivity

Training stability and runtime were observed to be sensitive to configuration choices such as batch size and evaluation frequency. Rather than maximizing throughput, the focus was placed on identifying stable configurations that minimize unnecessary computation while preserving reliable validation behavior.

9.3.3 Early Consideration of Precision and Memory

Precision-aware training strategies were considered from an early stage to assess their potential impact on memory consumption and iteration speed. While not fully integrated, these considerations informed later architectural planning.

9.4 Relation to Current and Future System Extensions

The findings obtained from this exploratory optimization study directly informed both the current system extensions and future development stages.

As the system evolved from a baseline classification pipeline into a multi-module AI-assisted lung nodule analysis framework, it became clear that computational cost is not limited to model training alone. The integration of segmentation, Grad-CAM-based explainability, anatomical localization, Turkish NLP-assisted reporting, preprocessing cache infrastructure, and UI-based inference introduced additional runtime and resource-related considerations.

In particular:

- The SegResNet-based segmentation pipeline introduced additional memory and computational overhead through preprocessing, ROI generation, and mask reconstruction.
- The Grad-CAM-based explainability module required additional activation map computation after inference.
- The anatomical localization algorithm added DICOM metadata processing and slice-level spatial analysis to the system workflow.
- The Turkish NLP and reporting modules introduced additional text-processing and structured-report generation layers.
- The UI-assisted inference workflow made preprocessing throughput and inference latency more important for user-facing system responsiveness.

These observations show that preprocessing cache reuse, intermediate data storage, and memory-

efficient processing are not only performance improvements but also essential requirements for system scalability.

Therefore, this section serves as a bridge between the initial baseline classification system and the current multi-module LungXai framework, while also providing engineering insights for future optimization and deployment-oriented improvements.

10. Limitations and Threats to Validity

Despite the promising results obtained in the current phase of the LungXai framework, several limitations remain that affect both the technical reliability and the clinical interpretability of the system. These limitations should be carefully considered when interpreting the experimental findings.

10.1 Limitations of Segmentation Accuracy

Although the SegResNet-based segmentation pipeline demonstrated the ability to learn meaningful lesion localization, the generated masks are not always perfectly aligned with the true anatomical boundaries of the nodules.

In some cases:

- partial mask coverage,
- boundary inaccuracies,
- or activation leakage into surrounding tissue

were observed during qualitative inspection.

Since the segmentation outputs are later used to support classification, Grad-CAM analysis, and localization workflows, segmentation inaccuracies may propagate to downstream modules and influence overall system behavior.

Therefore, the current segmentation module should be interpreted as a supportive localization mechanism rather than a fully reliable clinical segmentation system.

10.2 Limitations of Classification Reliability

The current classification pipeline is capable of producing meaningful benign–malignant discrimination signals; however, the generated predictions should not be interpreted as definitive clinical diagnoses.

In particular:

- high classification confidence does not necessarily guarantee pathological correctness,
- and high segmentation confidence may indirectly bias classification confidence due to shared anatomical focus regions.

For example, if the segmentation module strongly activates around a candidate region, the classification module may also produce a high-confidence prediction for the same area even when the underlying lesion characterization is uncertain.

Consequently, the current framework should be interpreted as a computer-aided decision support system rather than an autonomous diagnostic system.

10.3 Limitations of Explainability Analysis

Although Grad-CAM visualizations improved qualitative interpretability, the current explainability evaluation remains primarily visual and exploratory.

Several limitations remain:

- activation maps may occasionally spread beyond the true lesion region,
- small nodules may produce unstable heatmaps,
- and quantitative overlap validation between Grad-CAM outputs and segmentation masks has not yet been fully established.

Therefore, Grad-CAM outputs should be treated as supportive interpretability indicators rather than definitive evidence of model reasoning correctness.

10.4 Limitations of Anatomical Localization

The anatomical localization module currently relies on rule-based spatial estimation using:

- candidate center coordinates,
- DICOM metadata,
- and slice-position analysis.

Although this approach provides clinically meaningful lobe-level estimation in many cases, it does not yet perform:

- full anatomical lung segmentation,
- vessel-aware localization,
- or precise bronchopulmonary region analysis.

As a result, localization outputs should be interpreted as approximate anatomical estimations rather than exact clinical localization labels.

10.5 Limitations of the Turkish NLP and Reporting Module

The current Turkish reporting system is based on deterministic clinical rules combined with supportive BERTurk-assisted NLP analysis.

However, several limitations remain:

- the system cannot reliably determine nodule morphology categories such as solid or subsolid directly from imaging outputs,
- morphology-dependent Fleischner pathways are therefore only partially supported,
- and NLP outputs remain auxiliary rather than fully diagnostic.

In addition, the generated reports depend heavily on the availability and correctness of structured input parameters such as:

- nodule size,
- localization,
- and malignancy-related findings.

Therefore, the current reporting module should be interpreted as a supportive automated reporting prototype rather than a finalized clinical reporting system.

10.6 Dataset and Annotation Limitations

The LIDC-IDRI dataset contains:

- inter-reader variability,
- heterogeneous acquisition parameters,
- and ambiguous malignancy scores.

Although malignancy score aggregation and exclusion of indeterminate nodules were used to reduce label noise, annotation uncertainty remains an inherent limitation of the dataset.

Furthermore, some preprocessing and segmentation stages depend on metadata consistency between CT and SEG structures, making the pipeline sensitive to incomplete or inconsistent annotations.

10.7 Computational and Experimental Limitations

The project was developed under limited hardware conditions, which constrained:

- model complexity,
- batch size,
- and large-scale hyperparameter exploration.

Although preprocessing cache reuse and optimization-oriented strategies improved efficiency, more computationally demanding experiments such as:

- fully 3D segmentation,
- large-scale transformer training,
- and multitask learning

remain outside the scope of the current implementation.

As a result, the current framework should be interpreted as a modular research-oriented system rather than a fully optimized production-level clinical platform.

11. Future Work and Next Phase Plan

This section outlines the planned extensions and future development directions of the LungXai framework based on the currently implemented preprocessing, segmentation, classification, Grad-CAM, and nodule localization modules. The future work focuses on improving anatomical awareness, classification robustness, explainability reliability, localization precision, computational efficiency, and deployment readiness.

11.1 Segmentation-Guided Classification and Cross-Validation

The current system already includes a SegResNet-based segmentation pipeline that generates anatomically aligned ROI-based nodule masks from CT and SEG data. In the next phase, this segmentation output will be integrated more tightly with the classification pipeline to establish a segmentation-guided classification framework.

The primary goal of this integration is to reduce background influence and enforce spatial focus directly on the nodule region before malignancy prediction. In addition, segmentation outputs and classification predictions will be evaluated together through cross-validation-based consistency analysis. This

structure is expected to provide stronger anatomical supervision and improve the reliability of malignancy discrimination.

Future experiments will compare classification performance before and after segmentation-guided ROI refinement using metrics such as ROC-AUC, recall, F1-score, and specificity in order to quantify the contribution of segmentation to classification performance.

11.2 Explainable Artificial Intelligence (XAI)

The current LungXai framework already includes Grad-CAM-based interpretability support for visualizing the spatial regions influencing model predictions. Future work will focus on improving the reliability and quantitative evaluation of these explainability outputs.

In particular, future studies will investigate:

- quantitative overlap analysis between Grad-CAM heatmaps and segmentation masks using IoU and Dice metrics,
- confidence calibration for activation maps,
- attention-based visualization methods,
- and segmentation-restricted Grad-CAM generation.

These improvements aim to ensure that the model focuses on anatomically meaningful nodule regions instead of unrelated background structures.

11.3 Nodule Localization and Anatomical Region Estimation

The current system already includes a rule-based anatomical localization algorithm implemented through the `estimate_candidate_location` function. This module estimates the anatomical region of each candidate nodule using DICOM metadata and slice-level spatial analysis.

The algorithm:

- extracts center coordinates (`center_x`, `center_y`) for each candidate,
- calculates patient-space coordinates using `ImagePositionPatient` and `ImageOrientationPatient`,
- determines right/left lung positioning,
- estimates slice position within the lung volume using `lung_slice_start` and `lung_slice_end`,
- and assigns anatomical region labels such as:
 - RUL (Right Upper Lobe),
 - RML (Right Middle Lobe),
 - RLL (Right Lower Lobe),
 - LUL (Left Upper Lobe),

- LLL (Left Lower Lobe).

The module also returns additional information such as localization confidence and slice fraction values.

Future work will focus on improving localization precision through:

- segmentation-assisted lobe estimation,
- fully 3D anatomical localization,
- vessel and airway-aware positioning,
- and integration with UI-based visual nodule marking systems.

These improvements are expected to increase both clinical interpretability and user interaction quality.

11.4 Multitask Learning and Joint Prediction Frameworks

Future extensions will investigate multitask learning formulations in which the model jointly performs:

- malignancy classification,
- segmentation,
- Grad-CAM-based attention learning,
- and anatomical localization.

This approach is expected to improve generalization by forcing the network to learn anatomically meaningful shared representations instead of relying only on classification loss.

Joint prediction frameworks may also support:

- direct visual nodule marking inside the UI,
- automated report generation,
- and richer clinical decision-support outputs.

11.5 Model Acceleration and Efficiency-Oriented Studies

Further studies will focus on improving training and inference efficiency through:

- preprocessing cache reuse,
- precision-aware training,
- mixed precision inference,
- optimized data-loading pipelines,
- and memory-efficient segmentation processing.

Since segmentation and multitask learning significantly increase computational requirements, future optimization studies will focus on maintaining scalability under limited hardware resources.

Additional profiling studies may also analyze:

- GPU utilization,

- VRAM consumption,
- preprocessing bottlenecks,
- and inference latency.

11.6 Deployment-Oriented Structuring and Viewer Compatibility

Long-term development goals include restructuring the LungXai pipeline to operate in clinically compatible environments such as:

- OHIF Viewer,
- 3D Slicer,
- and DICOM-compatible medical imaging systems.

This integration will allow:

- segmentation masks,
- Grad-CAM heatmaps,
- classification predictions,
- and anatomical localization outputs

to be reviewed directly alongside original CT images within familiar medical imaging interfaces.

Future deployment-oriented work may also include:

- API-based inference services,
- real-time UI integration,
- cloud-compatible deployment structures,
- and report-generation pipelines integrated with hospital information systems.

12. Conclusion

This study presented LungXai, a modular and reproducible AI-assisted lung nodule analysis framework developed for chest CT-based malignancy assessment. Rather than proposing a monolithic end-to-end architecture from the beginning, the project followed a progressive development strategy focused on correctness, interpretability, modularity, and engineering robustness.

Throughout the current phase of the project, multiple interconnected system components were successfully implemented and evaluated, including:

- ROI-based preprocessing,
- UID-based CT-SEG matching,
- SegResNet-based segmentation,
- knowledge distillation-based classification,
- Grad-CAM-based explainability,
- anatomical nodule localization,
- Turkish NLP-assisted reporting,
- preprocessing cache optimization,
- and UI-assisted visualization.

The proposed preprocessing framework enabled anatomically aligned ROI generation and stable 2.5D sample construction from raw DICOM and SEG data. In addition, the segmentation pipeline demonstrated that meaningful lesion localization could be learned from ROI-based representations, while the knowledge distillation structure showed that lightweight 2.5D student models could successfully learn supportive volumetric information from higher-capacity teacher networks.

Experimental findings demonstrated that meaningful discrimination between benign and malignant nodules can be achieved using computationally efficient architectures. The experiments also revealed that ROC-AUC provides a more reliable evaluation criterion than accuracy for imbalanced medical classification problems and therefore should be prioritized during model selection and comparison.

Beyond classification performance, the study emphasized interpretability and system-level integration. The Grad-CAM module enabled qualitative inspection of model attention behavior, while the anatomical localization module provided clinically meaningful region estimation using DICOM metadata and slice-level spatial analysis. In addition, the Turkish NLP-assisted reporting system demonstrated how image-based AI outputs can be combined with structured clinical workflows and deterministic reporting logic.

From an engineering perspective, the project established a modular pipeline architecture separating

preprocessing, segmentation, classification, explainability, localization, reporting, and UI layers. This separation improved reproducibility, maintainability, and scalability while enabling independent development and evaluation of individual system components.

Furthermore, the exploratory optimization study highlighted important computational considerations related to preprocessing overhead, memory usage, inference latency, and multi-module integration. The introduction of .npz-based preprocessing cache infrastructure demonstrated that efficient data reuse becomes increasingly important as segmentation, explainability, localization, and UI-driven inference pipelines are added to the system.

Overall, the current LungXai framework demonstrates that combining anatomically aware preprocessing, interpretable deep learning, localization support, and clinically oriented reporting within a unified modular architecture is both feasible and extensible. The developed system establishes a strong engineering and experimental foundation for future deployment-oriented and clinically scalable AI-assisted lung nodule analysis systems.

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